

Beyond One-Step Accuracy: Cell-to-Cell Variability, Transfer, and Calibrated Uncertainty in Deployment-Oriented Health Prediction for Large-Format LiFePO₄ Cells

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ABSTRACT

Accurate state-of-health (SOH) forecasting is important for battery management, yet most machine-learning studies are evaluated on small laboratory cells with rolling one-step-ahead prediction, where the model is continuously supplied with the latest observed SOH. This study examines whether such benchmarks reflect the challenges of large-format battery deployment. We analyse an 18-month dataset of 20 large-format 50 Ah LiFePO₄ prismatic cells tested under three operating conditions, together with several external battery datasets. Substantial cell-to-cell variation is observed even among nominally identical cells, while a comparison of six deep-learning models shows that conventional one-step forecasting is close to saturation: a simple linear autoregressive model performs as well as or better than the deep models, and cell identity explains substantially more prediction-error variance than model choice.

These findings motivate a deployment-oriented framework that targets capabilities beyond one-step accuracy. The proposed hybrid model combines LFP pretraining, an early-life fingerprint encoder over the first 30 cycles, target-cell fine-tuning, and uncertainty estimation. Without target-cell fine-tuning, fingerprint-conditioned transfer reduces RMSE by 23% relative to training from scratch, and subsequent fine-tuning increases the average reduction to 51% over the same scratch baseline. The framework is further evaluated for uncertainty estimation, early identification of fast-degrading cells, cross-dataset transfer, and remaining-useful-life forecasting. Early-life voltage-curve features flag cells at risk of premature degradation with an AUC of 0.78, and the RUL model reaches a median error of 49 cycles at the midlife forecasting point. The proposed hybrid is not intended to win the already-saturated one-step benchmark; its value is transfer to a new cell, uncertainty estimation, early screening, and long-term forecasting. Overall, the main challenge in large-format battery prognostics is not further reducing already-small one-step errors, but developing models that adapt across cells, quantify uncertainty, identify abnormal degradation early, and provide reliable long-term forecasts.

1. Introduction

The accurate estimation of battery state of health (SOH) is a prerequisite for the safe and efficient operation of lithium-ion batteries in electric vehicles (EVs), grid-scale energy storage, and consumer electronics [1]. SOH is the ratio of a cell's present usable capacity to its original rated capacity; as a cell is charged and discharged over hundreds of cycles this ratio falls, and the point at which it drops too far sets the battery's usable lifetime and dictates when it must be replaced [2]. Predicting how SOH will decline, ideally early and for each individual cell, is therefore central to managing cost, safety, and warranty risk in every large battery deployment.

Abbreviations: AUC, Area Under the ROC Curve; BMS, Battery Management System; CNN, Convolutional Neural Network; DE, Differential Evolution; EOL, End of Life; EV, Electric Vehicle; FP, Fingerprint; FT, Fine Tuning; LFP, Lithium Iron Phosphate; LCO, Lithium Cobalt Oxide; LOOCV, Leave-One-Out Cross-Validation; LSTM, Long Short-Term Memory; MAE, Mean Absolute Error; MC, Monte Carlo; NCA, Nickel Cobalt Aluminium; NMC, Nickel Manganese Cobalt; PI, Prediction Interval; PT, Pretraining; RMSE, Root Mean Square Error; RUL, Remaining Useful Life; SIT, Singapore Institute of Technology; SOH, State of Health; ZS, Zero-Shot.

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Over the past decade, deep learning has produced increasingly capable SOH prediction models, progressing from standard long short-term memory (LSTM) networks to attention-augmented variants [3] and models whose hyperparameters are tuned by metaheuristic search [4, 5]. Yet almost all of these methods are developed and benchmarked in the same narrow setting: small cylindrical 18650 cells of 1 to 3 Ah (LCO, NMC, or NCA chemistry) cycled at constant current in the laboratory. The NASA Prognostics Repository [6], and in particular its cells B05, B06, B07, and B18, has been the field's de facto benchmark since 2007, appearing in roughly 30.8% of all published SOH and remaining-useful-life (RUL) studies [7, 8].

This near-exclusive reliance on small laboratory cells creates a gap that only becomes visible at deployment. The batteries that actually run electric vehicles and grid storage are large-format prismatic or pouch cells of 10 to 200 Ah [9], operated across a wide range of temperatures and charge and discharge rates. Crucially, cells that are nominally identical, built to the same design on the same production line, can age very differently because of small, unavoidable variations introduced during manufacturing [2]. This cell-to-cell variability, of which manufacturing spread is a likely contributor, is largely absent from the hand-picked, smoothly degrading cells that populate standard benchmarks, yet it is

precisely what a manufacturer or fleet operator must contend with: warranties, safety margins, and replacement schedules all depend on predicting the life of each individual cell, not an average one. Just how severe this variability is on large-format cells, and whether it, rather than the choice of algorithm, sets the ultimate limit on prediction, remains unknown.

A second, more subtle problem lies in how the field measures success. Almost all of this literature ranks methods by one-step-ahead error: given a cell's observed history, predict its SOH one cycle into the future. On the smooth, slowly changing trajectories of large-format cells, however, this is barely harder than assuming that the next cycle simply repeats the last, which raises a basic question: can a metric so close to a trivial baseline actually tell one method apart from another? If it cannot, then a leaderboard built on it is uninformative, and predictors need to be judged by something that matters in the field instead, the practical capabilities they enable: predicting a brand-new cell before it has aged, attaching trustworthy confidence bounds to each forecast, flagging defective cells early, and forecasting remaining useful life.

The objective of this paper is to close this gap between benchmark practice and deployment reality. Using an 18-month dataset of 20 large-format LiFePO₄ prismatic cells (50 Ah) cycled under three temperature and C-rate conditions [10], we set out to establish how far cell identity, rather than model choice, governs prediction on industrial-format cells, and to determine what a prediction method must provide to be useful once these cells are deployed. In place of a single accuracy score, we take as our goal a model that can predict an unseen cell from only its earliest cycles, report and correct its own uncertainty, remain reliable when the available training data come from a different battery chemistry, and extend to the related deployment tasks of early defect screening and remaining-useful-life forecasting.

The remainder of this paper is organised as follows. Section 2 reviews the related literature. Section 3 states the research gaps and summarises the contributions of this work. Section 4 describes the datasets, the DE-LSTM baseline, and the evaluation protocol. Section 5 characterises the deployment challenges empirically on the SIT cells. Section 6 presents the proposed hybrid model. Section 7 reports its evaluation, including the ablation study and the cross-dataset validation. Section 8 shows how the manufacturing-defective cells can be detected early from the discharge voltage curve. Section 9 extends the hybrid to remaining-useful-life forecasting. Section 10 discusses implications for deployment, and Section 11 concludes.

2. Literature Review

This section reviews the prior work most relevant to this study, spanning SOH prediction methods, large-format and LFP cells, transfer learning, cell-to-cell variability, and uncertainty quantification.

2.1. SOH Prediction Methods

Data-driven SOH prediction has evolved from simple regression models to sophisticated deep-learning architectures. Recurrent neural networks, particularly LSTM variants, have emerged as the dominant approach due to their ability to model long-range temporal dependencies in discharge capacity sequences [11]. Convolutional architectures form a second major family: deep CNNs with ensemble and transfer learning have been applied to capacity estimation [12, 13], building on early demonstrations that deep networks can estimate battery state directly from operational measurements [14].

Attention mechanisms have been incorporated to allow models to weight different timesteps in the input window [15]. Qu et al. proposed PA-LSTM, which augments an LSTM encoder with an attention layer and uses particle swarm optimisation (PSO) for hyperparameter tuning and model pretraining, with incremental learning for online SOH monitoring; PA-LSTM outperformed vanilla LSTM, recurrent neural network (RNN), and relevance vector machine (RVM) baselines on NASA cells [3]. Subsequent work replaced PSO with other metaheuristics for LSTM hyperparameter optimisation: genetic algorithm (GA-LSTM) [4] and differential evolution [16] (DE-LSTM) [5] both achieved significant RMSE reductions on the same NASA cells (B05, B06, B18), retained from PA-LSTM to enable direct comparison.

A common thread across these methods is the use of *incremental (walk-forward) learning*: the model is trained on the first $p\%$ of a cell's SOH trajectory and evaluated on the remainder, simulating online deployment. This protocol is adopted throughout the present work. The proposed hybrid uses the same walk-forward protocol but introduces a fundamentally different use of early-cycle data: rather than fitting the observed sequence to continue it, a convolutional encoder summarises the *shape* of the first 30 cycles into a cell-identity embedding that initialises the LSTM hidden state before fine tuning begins. This places each cell in the pretrained population's manufacturing distribution, discriminating between cell types even when early trajectory magnitudes appear similar to a sequence-fitting model.

2.2. Large-Format and LFP Batteries

Most published SOH prediction studies use 18650-format cylindrical cells (LCO, NMC, NCA, or LFP chemistry, 1 to 3 Ah). Large-format prismatic and pouch cells are the dominant format in stationary storage and electric heavy vehicles [17], yet they are substantially underrepresented in the prediction literature. Key differences include: non-uniform current distribution across the electrode area [9]; larger absolute capacity changes per SOH unit (a 1% SOH change in a 50 Ah cell = 0.5 Ah vs. 0.02 Ah in a 2 Ah cell); and different thermal management requirements.

LiFePO₄ (LFP) cells are preferred for stationary storage due to their flat discharge plateau, long cycle life, and superior thermal stability [17]. The flat voltage plateau of LFP makes capacity based SOH estimation more robust than

voltage based methods, motivating the discharge-capacity incremental learning approach used in this work.

2.3. Transfer Learning for Battery SOH

Transfer learning (pretraining a model on a source dataset and adapting it to a target domain) has attracted growing interest for battery prognostics [18]. Approaches include fine tuning pretrained networks, domain-adversarial training, and meta-learning [19]. Yu et al. [20] addressed cross-domain SOH estimation across heterogeneous small-cell datasets (NASA and CALCE), fusing the most similar source domains through a similarity network to maintain accuracy when target-cell data are scarce. Ye and Yu [21] applied domain-adversarial transfer learning to align feature distributions between source and target cells for SOH estimation. Closest to the present work, Ma et al. [22] achieve personalised, real-time prediction by fine tuning a pretrained model on each cell's most recent cycles. The fingerprint encoder proposed here uses early-life data differently: it treats the first 30 cycles as a static shape that sets the LSTM initial state before fine tuning, rather than fine tuning on those cycles directly (Section 6.4).

A key practical constraint is chemistry: LFP cells have a fundamentally different degradation mechanism (lithium plating, phase transition stress) compared to layered-oxide cells such as LiCoO₂ (LCO) and NMC (transition metal dissolution, solid electrolyte interphase (SEI) growth) [2]. Whether a model pretrained on one chemistry can transfer to another is an open question addressed in this work through explicit comparison of chemistry matched (LFP→LFP) and chemistry mismatched (LFP→LCO) transfer.

2.3.1. Cross-Dataset and Foundation-Model Transfer

A recent line of work asks the same question at corpus scale, pretraining a single model over several public collections and then adapting it to a target. Zhang et al. [23] are closest to our remaining-useful-life setting: they forecast capacity trajectories with a time-series foundation model across four public benchmarks, observing the first 30% of life and predicting the remainder, under a protocol that separates zero-shot inference from prefix-adapted and source-adapted variants. Their central result is that source adaptation helps most when source and target come from the same collection, while cross-dataset transfer degrades accuracy under distribution shift.

That finding is independent corroboration of the domain-alignment effect we report in Section 7, reached from a different direction: they vary the dataset at fixed chemistry, whereas we vary the chemistry and hold the degradation mode fixed. We adopt their vocabulary for the distinction that matters here, describing our encoder as fingerprint-conditioned, that is, prefix-adapted without target-cell fine tuning, rather than as zero-shot in the looser sense the term often carries.

Our contribution is differential rather than unprecedented. This body of work is built almost entirely on small cylindrical cells of 1 to 3 Ah. What we add is a large-format setting, 50 Ah prismatic cells with a per-cell thermal record,

and an explicit test of whether calibrated uncertainty, not just point accuracy, survives the transfer.

2.4. Cell-to-Cell Variability

Cell-to-cell variability arising from manufacturing tolerances is well-documented in the electrochemical literature [2] but rarely quantified in prediction studies. Dubarry et al. [24] showed that even cells from the same batch exhibit measurable differences in electrode loading and electrolyte fill that propagate to divergent long-term trajectories. Severson et al. [18] observed that early-cycle statistics (first 100 cycles) correlate with cycle life, motivating the fingerprint approach adopted here.

Few prior studies quantify cell-to-cell variability across identical large-format LFP cells or examine how cross-cell pooling interacts with this variability, the *two-sided failure mode* we characterise in Section 5.2.

2.5. Uncertainty Quantification

Point-estimate SOH predictions are insufficient for BMS deployment, where decisions under uncertainty require calibrated prediction intervals. Bayesian neural networks [25] and Monte Carlo Dropout [26] have been applied to battery SOH uncertainty estimation [27], but calibration is rarely verified against held-out data, and never across a chemistry mismatch. Conformal prediction provides distribution free coverage guarantees [28], an attractive property for safety-critical battery management. In this work we use MC-Dropout and quantify its empirical coverage, revealing a chemistry-alignment effect in which calibration holds only when the pretraining and deployment chemistries match.

3. Research Gap and Contributions

This section sets out the gaps left open by prior work and summarises how this study addresses them.

3.1. Research Gap

Despite substantial progress in data-driven SOH prediction, three critical gaps remain unaddressed in the existing literature:

1. **Cell-to-cell variability is not modelled.** Published benchmarks typically use hand-picked cells with moderate, smooth degradation trajectories. Cell-to-cell variance, to which manufacturing tolerance is a likely contributor, can produce 10× differences in degradation rate under identical conditions, is systematically understudied. Most transfer and fine tuning methods adapt to a target cell either by aligning feature distributions across batteries at the population level [21] or by fine tuning a shared model on the target cell's own recent data [22]. Neither extracts an explicit early-life shape signature that locates a cell within the manufacturing distribution of the pretrained population before adaptation, so a cell whose early trajectory is ambiguous (Section 5.2) cannot be distinguished from an average cell.

2. **The standard benchmark protocol has never been validated.** Nearly all published comparisons, including our own prior work [4, 5], rank methods by one-step-ahead RMSE on small cylindrical 18650 cells (with NASA cells B05, B06, B07, and B18 the de facto standard since 2007 [6, 7]). Whether this protocol can actually distinguish methods, that is, whether the reported differences exceed trivial-baseline performance and the run-to-run variance of training itself, has never been tested, and neither has its behaviour on large-format (≥ 10 Ah) cells under industrially representative conditions.
3. **Uncertainty calibration is chemistry-dependent.** Bayesian approximations and conformal prediction, including MC-Dropout and Bayes by Backprop (BBB), provide well-studied uncertainty estimates [26, 28] and have been applied to battery SOH [27], but typically within a single chemistry. Whether these estimates remain calibrated under a chemistry mismatch between pretraining and deployment data has not been investigated, a critical gap for practical BMS deployment where pretraining data may be available only for a different battery chemistry.

This study addresses these three gaps using the SIT dataset [10], our previously published large-format campaign of 20 LiFePO₄ prismatic cells (50 Ah, 3.2 V) cycled over 18 months under three operating conditions. As the baseline we use our previously published DE-LSTM [5], a differential-evolution-optimised LSTM with leading accuracy on the NASA 18650 benchmark, alongside five further deep baselines and two trivial baselines. Section 5 quantifies how severely the first two gaps, cell-to-cell variability and benchmark saturation, manifest on these cells; Sections 6 and 7 present and validate the proposed hybrid, whose value lies in deployment capabilities rather than one-step rank; and a cross-dataset evaluation establishes the chemistry-alignment effect, in which the model's uncertainty estimates stay calibrated only when pretraining and deployment chemistries match.

3.2. Contributions

This paper makes the following key contributions:

1. **Large-format benchmark with trivial-baseline control:** the first systematic evaluation of six deep single-cell predictors, including our prior DE-LSTM, on 50 Ah LiFePO₄ prismatic cells, together with linear-autoregressive and curve-fitting controls that expose the saturation of the one-step protocol (Section 5.1).
2. **Cell-to-cell variability quantification:** Across ten nominally identical G1 cells, we show that although each cell's own degradation trajectory is highly predictable, the cells differ so much from one another that this spread between cells, not the algorithm, sets the error floor. A variance decomposition across six predictors attributes 25% to 58% of error variance to the cell and only $\sim 15\%$ to the method, and cross-cell

pooling exhibits a two-sided failure mode: it rescues the fast-degrading cells but worsens others by up to $2\times$ (Section 5.2).

3. **Benchmark-saturation finding:** We show that the field's standard one-step-ahead protocol cannot rank methods on data-rich large-format cells: a linear autoregressor attains the lowest error of all evaluated models, inter-method differences fall within training-run variance, and apparent cross-condition effects are dominated by the quality of individual training runs (Sections 5.1 and 5.3).
4. **Hybrid transfer model with ablation:** We propose a three-phase model (MIT LFP pretraining, convolutional fingerprint encoder, MC-Dropout uncertainty) and conduct a full ablation study isolating each component's contribution. The hybrid predicts new cells from 30 cycles with no target-cell fine-tuning (23% better than scratch) and achieves a 51% mean RMSE improvement over scratch with fine tuning, on all 17 of 17 test cells; it is the only evaluated method that quantifies uncertainty, with prediction intervals that are recalibrated by grouped conformal inference (Section 7.1). We additionally characterise pretraining-run variance, show that source-domain criteria cannot detect poorly transferring pretraining runs, and introduce a calibration-cell checkpoint-selection protocol that makes the pipeline reliably reproducible (Section 6.3).
5. **Chemistry-alignment effect with multi-dataset validation:** We demonstrate that MC-Dropout uncertainty calibration transfers between datasets only when pretraining and deployment chemistries are matched (LFP $\rightarrow$ LFP: 80.8% coverage; LFP $\rightarrow$ LCO: 43.6% coverage). A ten-scenario cross-dataset experiment spanning three LFP datasets (MIT, SIT, LISHEN 40 Ah) and one LCO dataset confirms this effect and shows that fine tuning is the dependable adaptation mechanism across chemistries, while fingerprint conditioning without fine-tuning provides its distinctive value before any adaptation is possible, yielding a practical guideline for transfer learning in BMS applications (Sections 7.2 and 10).
6. **Early screen for manufacturing-defective cells:** We show that no scalar health indicator separates the defective cells within a group, yet the early-cycle discharge voltage-curve descriptor $\Delta Q(V)$ detects eventual early death at an AUC of 0.78 on the SIT cells, reproduced on the MIT corpus, about 50 cycles before the capacity trajectory reveals the defect (Section 8).
7. **Full-coverage probabilistic RUL forecasting:** We extend the hybrid from one-step SOH to remaining-useful-life forecasting with a direct multi-horizon decoder and a 155-cell multi-dataset LFP foundation corpus, forecasting every cell that reaches end of life, including the catastrophic cells that recursive baselines abstain on, at a median error of 49 cycles at the midlife anchor with conformally calibrated intervals (Section 9).

4. Dataset and Experimental Methodology

This section describes the datasets used in this study, the DE-LSTM baseline against which the hybrid is developed, and the evaluation protocol shared by all experiments.

4.1. Datasets

4.1.1. SIT LiFePO₄ Dataset (Primary)

The SIT dataset [10] comprises 20 commercial LiFePO₄ prismatic cells (Narada Power Source, 50 Ah nominal, 3.2 V) cycled in the Singapore Institute of Technology Battery Laboratory over 18 months. Cells are organised into three groups differing in temperature and C-rate:

- **G1** (10 cells, IDs 001-1 to 001-8, 101-1, 101-3): ambient temperature (≈ 25), 1 C charge/discharge, deep cycling. All cells share nominally identical specifications and protocol, so observed differences are most plausibly dominated by manufacturing tolerances, although cycler-channel differences, contact resistance, minor temperature variation, testing interruptions and production batch may also contribute.
- **G2** (4 cells, IDs 002-1 to 002-4): 40 thermal chamber, 1 C rate.
- **G3** (6 cells, IDs 002-5, 002-7, 003-1, 003-3, 003-5, 003-7): 40 thermal chamber, 2 C rate.

SOH is computed from discharge capacity as $\text{SOH}(k) = Q_k/Q_1$, where Q_k is discharge capacity at cycle k and Q_1 is the first-cycle capacity. Table 1 summarises cycle counts and final SOH for all 20 cells, where SOH_{end} is the final-cycle SOH. Cell 003-5 records $\text{SOH} > 1$, a capacity-recovery artefact arising at low C-rate after elevated-temperature cycling; it is retained in the analysis but treated with caution.

G1 provides the cell-to-cell variability experiment (Section 5.2); G1/G2/G3 provide the cross-condition transfer experiment (Section 5.3).

4.1.2. MIT LFP Dataset (Pretraining)

The Severson et al. [18] dataset provides 124 LFP/graphite 18650 cells (1.1 Ah) cycled at Stanford University and Toyota Research Institute at various C-rates and charging protocols. Of these, 49 cells have complete cycle-level capacity data in our processing; the *pretraining corpus* comprises the 40 batch-1 cells (batch1_cell000 to 039). This choice is empirical: in a corpus-sensitivity test, extending the corpus with the nine remaining complete-data cells (including the two batch-3 cells with $\leq 9\%$ total fade) collapsed the next-cycle pretraining objective toward a trivial persistence solution (validation MSE $3.4 \times 10^{-3} \rightarrow 1 \times 10^{-5}$) and degraded downstream transfer; the 40-cell corpus is therefore retained, and the full corpus-sensitivity results are provided in the released repository. We emphasise that this is a property of the pretraining objective, not a fragility of the deployed model: the hybrid is evaluated on, and performs well on, slowly fading target cells (e.g. the G2 cells with final SOH ≈ 0.9). The early-life correlation analysis of Table 7 uses

Table 1

SIT cell inventory across three operating conditions.

Group	Cell	Cycles	SOH _{end}	Condition
G1	001-1	876	0.767	25°C, 1 C
	001-2	764	0.878	
	001-3	788	0.493	
	001-4	878	0.709	
	001-5	684	0.885	
	001-6	876	0.854	
	001-7	876	0.885	
	001-8	764	0.067	
	101-1	864	0.696	
	101-3	697	0.118	
G2	002-1	564	0.924	40°C, 1 C
	002-2	572	0.888	
	002-3	561	0.915	
	002-4	560	0.660	
G3	002-5	689	0.913	40°C, 2 C
	002-7	695	0.564	
	003-1	603	0.461	
	003-3	693	0.559	
	003-5	567	1.075 [†]	
	003-7	694	0.587	

[†] Capacity recovery artefact; included in analysis with caution.

the 47 genuine degraders (final SOH < 1) among all 49 complete-data cells, excluding the two capacity-recovery artefacts. Pretraining on chemistry matched LFP cells is a deliberate design choice motivated by the chemistry and domain alignment effect evaluated in Section 10.

4.1.3. External Datasets

Four further datasets support the cross-chemistry test and the transfer and RUL experiments. The NASA Prognostics Center of Excellence dataset [6] provides 26 lithium cobalt oxide (LCO) 18650 cells of about 2 Ah cycled at several ambient temperatures under constant-current discharge; because its chemistry differs from the LFP target cells, it serves as the cross-chemistry generalisation test. The LISHEN dataset [29] provides 3 large-format 40 Ah LFP prismatic cells cycled at 2 C over a 20% depth-of-discharge window, used as a third, independent LFP source in the cross-dataset transfer matrix. The HUST [22] and Sandia (SNL) [30] collections provide 77 and 29 LFP 18650 cells respectively; together with the 49 MIT cells they form the 155-cell LFP foundation corpus that pretrains the RUL decoder (Section 9.2). Table 2 consolidates the chemistry, format, capacity, cell count, operating conditions, and role of every dataset used in this study.

4.2. Baseline: DE-LSTM

DE-LSTM [5] is the strongest model from our prior work and serves as the baseline against which the hybrid is developed. It is a plain LSTM network whose hyperparameters (hidden units H , look-back window W , learning rate η) are optimised by differential evolution (DE) [16] using 2-fold walk-forward cross-validation as the fitness function; the batch size is set deterministically to $\max(32, H)$ rather than

Table 2

Datasets used in this study.

Dataset	Chemistry	Format	Cap. (Ah)	Cells	Conditions	Role in this study
SIT (primary) [10]	LFP	Prismatic	50	20	25 to 40 °C, 1 to 2 C	Target cells: testing and checkpoint selection
MIT/Severson [18]	LFP	18650	1.1	40	Fast charge, multi-protocol	Base-hybrid pretraining (training)
NASA PCoE [6]	LCO	18650	2	26	Varied temperature, constant current	Cross-chemistry test (testing)
LISHEN [29]	LFP	Prismatic	40	3	2 C, 20% depth of discharge	Third LFP source, cross-dataset transfer
HUST [22]	LFP	18650	1.1	77	Multi-rate discharge	RUL foundation corpus (training)
SNL (Sandia) [30]	LFP	18650	1.1	29	Varied temperature, rate, DOD	RUL foundation corpus (training)

searched. The architecture is:

$$\hat{y}_t = \mathbf{W}_o h_t + b_o, \quad h_t, c_t = \text{LSTM}(x_{t-W:t}, h_{t-1}, c_{t-1}) \quad (1)$$

where $x_{t-W:t}$ is the SOH look-back window of length W , h_t and c_t are the LSTM hidden and cell states at cycle t , $\mathbf{W}_o$ and b_o are the output-layer weight matrix and bias, and $\hat{y}_t$ is the predicted next-cycle SOH.

For the SIT baseline evaluation (Section 5.1), DE is run independently for each cell and each training split, with the search space $H \in \{8, 16, 32, 64, 128, 256\}$, $W \in [3, 30]$, $\eta \in [10^{-4}, 0.05]$. The DE search uses a population of 3 per parameter and a maximum of 8 generations (polish=False, tol = 10^{-4}), on the order of 80 candidate evaluations per cell and split, and is itself run under a single fixed seed. The selected hyperparameters are then re-evaluated under two independent random seeds, and the mean RMSE is reported.

4.3. Evaluation Protocol

All experiments use three training splits: 30%, 50%, and 70% of each cell's SOH trajectory, with the remainder held out for evaluation. Because each split is defined as a fraction of a cell's already-completed trajectory, these splits are retrospective (they presuppose the cell's total lifetime). Prediction is performed in a rolling one-step-ahead manner: at each held-out cycle the model receives the *observed* previous W SOH values and predicts the next one, and predicted values are never fed back as inputs (i.e., we report one-step-ahead prediction, not recursive multi-step forecasting). Hybrid-model evaluations are averaged over five random seeds per cell; single-cell baseline methods (Simple, PA-, GA-, DE-LSTM, TCN, Transformer) are averaged over two seeds after their hyperparameter search, which itself runs under a single fixed seed. Summary tables report the mean RMSE with its standard deviation across the cells in each group. Improvement over scratch LSTM is computed per cell and then averaged, so a reported improvement is the mean of the per-cell percentage RMSE reductions rather than the percentage change between the group-mean RMSEs.

4.3.1. Data Partitioning and Leakage

Within each target cell the training and evaluation partitions are contiguous in cycle order: training windows are drawn from the first $p\%$ of the trajectory and evaluation windows from the remainder, so no held-out SOH value is ever used as a training target. The first evaluation window is seeded with the last W *observed* training cycles as its input; this is inherent to one-step-ahead prediction and exposes

no held-out target. For pretraining, the corpus cells are partitioned into training and validation folds *by whole cell, before any windowing*, so no cell contributes windows to both partitions and the early-stopping signal is measured only on cells the model has not seen. An earlier version of this pipeline pooled the windows in cell order and held out the final 10% of the pooled array, which allowed the single cell straddling that boundary to contribute mutually overlapping windows to both sides. That split was corrected to the cell-disjoint form described here, and every result reported in this paper comes from the corrected pipeline. Finally, since a cell of length T yields $T - W$ windows, cells with longer trajectories contribute proportionally more pretraining samples: the pooled corpus is sample-weighted by trajectory length rather than balanced per cell.

4.3.2. Calibration Cells and Test Cells

Three SIT cells, the numerically first cell of each condition group (001-1, 002-1, 002-5), designated *a priori*, are reserved as *calibration cells* for pretraining-checkpoint selection (Section 6.3). All reported SIT results are computed on the remaining **17 test cells**; the calibration cells never contribute to any reported metric.

4.3.3. Reproducibility

All experiments run with deterministic GPU kernels (TensorFlow op-determinism, full float32) and complete seeding of the Python, NumPy, and TensorFlow generators, so every reported number is bit-for-bit reproducible on the same hardware and software stack. The code, the full pipeline, and the selected pretrained checkpoint are permanently archived on Zenodo [31]. Statistical significance of pairwise comparisons uses the Wilcoxon signed-rank test [32] at $\alpha = 0.05$. This non-parametric test is preferred over a paired t -test because per-cell RMSE distributions are skewed by outlier cells (e.g., catastrophic degraders such as cell 001-8), and the Wilcoxon test does not assume normality. For the hybrid model, empirical 95% prediction-interval (PI) coverage (fraction of test points within the predicted interval) and mean PI width are additionally reported. These MC-Dropout intervals are approximate prediction intervals, not calibrated prediction intervals in the conformal sense.

4.3.4. Model Configurations and Tuning Budget

Table 3 reports every model's architecture, tuning budget, and training configuration exactly as run. The three

Table 3

Model configurations and training budgets, as run (per cell and split).

Model	Architecture	Tuning budget	Training configuration
Simple LSTM	LSTM(64) → Dense(1)	none (fixed)	$W=20$, $\eta=10^{-2}$, batch 64, 200 epochs, early stop patience 10
PA-LSTM	LSTM → attention → pooling → Dense(1)	PSO, ≈ 80 evals	searched: $H \in \{8 \dots 256\}$, $W \in [3, 30]$, $\eta \in [10^{-4}, 5 \times 10^{-2}]$; batch $\max(32, H)$, 200 epochs, patience 10
GA-LSTM	LSTM → Dense(1)	GA, 9×9 evals	searched, as PA-LSTM
DE-LSTM	LSTM → Dense(1)	DE, ≈ 80 evals	searched, as PA-LSTM
TCN	3 residual dilated causal conv blocks (64 filters, $k=3$, dilations 1, 2, 4)	none (fixed)	$W=20$, $\eta=10^{-2}$, batch 64, 200 epochs, patience 10
Transformer	1 encoder block ($d_{\text{model}}=64$, 4 heads, feed-forward 128)	none (fixed)	$W=20$, $\eta=10^{-3}$, batch 64, 200 epochs, patience 10
Scratch LSTM	LSTM(64) → Dropout(0.2) → Dense(1)	none (fixed)	$W=20$, $\eta=10^{-3}$, 100 epochs, patience 10
Hybrid (ours)	CNN fingerprint (Conv1D 32 and 16, $k=3$, global average pooling, Dense 16) → LSTM(64) initialised with (h_0, c_0) → MC-Dropout(0.2) → Dense(1); 21 089 parameters	none (fixed)	pretrain: $\eta=10^{-3}$, batch 64, 100 epochs, patience 12; fine tune: $\eta=5 \times 10^{-4}$, 50 epochs, patience 8; 50 MC-Dropout passes

search-based LSTMs (PA-, GA- and DE-LSTM) each receive a per-cell, per-split hyperparameter search of comparable size (of the order of 80 candidate evaluations), whereas the Simple LSTM, the TCN and the Transformer share a common fixed configuration. We note this asymmetry explicitly: the non-recurrent baselines are not hyperparameter-optimised, so their weaker performance should be read as evidence that a richer backbone is not what these data reward rather than as a tuned upper bound. The proposed hybrid likewise receives *no* per-cell search, and so competes against the searched LSTM baselines from a single fixed configuration.

5. Problem Characterisation: Limitations of Standard Methods

Before introducing the hybrid model, we use the SIT cells to characterise the problem landscape: we establish the DE-LSTM baseline, show that the standard one-step evaluation protocol saturates on these data, quantify the cell-to-cell variability that limits pooled prediction, and examine how training-run variance confounds cross-condition comparisons.

5.1. Baseline Performance on the SIT Cells

Figure 1 shows the SOH trajectories for all 20 cells. Within G1 (identical conditions), final SOH spans 0.067 to 0.885 (a 13.2× ratio), illustrating the scale of cell-to-cell variability. G2 cells (elevated temperature) show consistently high residual SOH (0.660 to 0.924), suggesting the 40 chamber protocol did not accelerate degradation within the test duration. G3 cells (elevated temperature and 2C rate) show more diverse outcomes (0.461 to 1.075), including two cells with moderate to severe degradation.

Table 4 reports DE-LSTM RMSE across all 20 SIT cells at 30%, 50%, and 70% training splits; values are mean $\pm$ std across the cells in each group, averaged over two evaluation seeds.

Two patterns are immediately evident. First, within G1, per-cell RMSE still varies $\sim 5\times$ (0.004 to 0.021 at the 50% split) despite identical conditions (Figure 2), and the worst cells at every split are the catastrophic degraders and the recovery artefact, whose early history is unrepresentative of their future. Second, once half a cell’s history is available the per-cell errors are small (~ 0.01) and nearly uniform across operating conditions, a first hint that the one-step task itself is close to its floor, which the next comparison makes explicit.

5.1.1. Comparison with Established Methods

To confirm that this error floor reflects the dataset rather than a weakness of DE-LSTM specifically, we compare six single-cell predictors on the 17 test cells under the identical protocol: four LSTM-family methods (a fixed-hyperparameter Simple LSTM, the attention-based PA-LSTM [3], GA-LSTM [4], and DE-LSTM [5]) and two modern non-recurrent architectures: a temporal convolutional network (TCN, dilated causal convolutions) and a Transformer encoder, each under a fixed configuration matched to the Simple-LSTM protocol. Table 5 reports the mean RMSE over the test cells at each split.

Table 5 shows that the one-step protocol is saturated on these data: a **20-cycle linear autoregressor attains the lowest error at every split**, on par with or ahead of all six deep models, and even a **memoryless persistence predictor** ($\hat{y}_t = y_{t-1}$, RMSE 0.013 at 50%) matches the best deep model (DE-LSTM, 0.0118) and beats the proposed hybrid (0.0156) on this metric. Among the deep models themselves no consistent ranking emerges (PA-LSTM leads at 30%, DE-LSTM at 50% and 70%, in sharp contrast to the NASA benchmark where DE-LSTM reduces RMSE by 85 to 91% over PA-LSTM and GA-LSTM [5]). This is a property of the task rather than a weakness of any one model, and it is precisely why, on large-format cells, a predictor must be judged by the capabilities it provides rather than by its one-step rank, the case the rest of this paper makes for the proposed hybrid. The explanation is that rolling one-step-ahead prediction

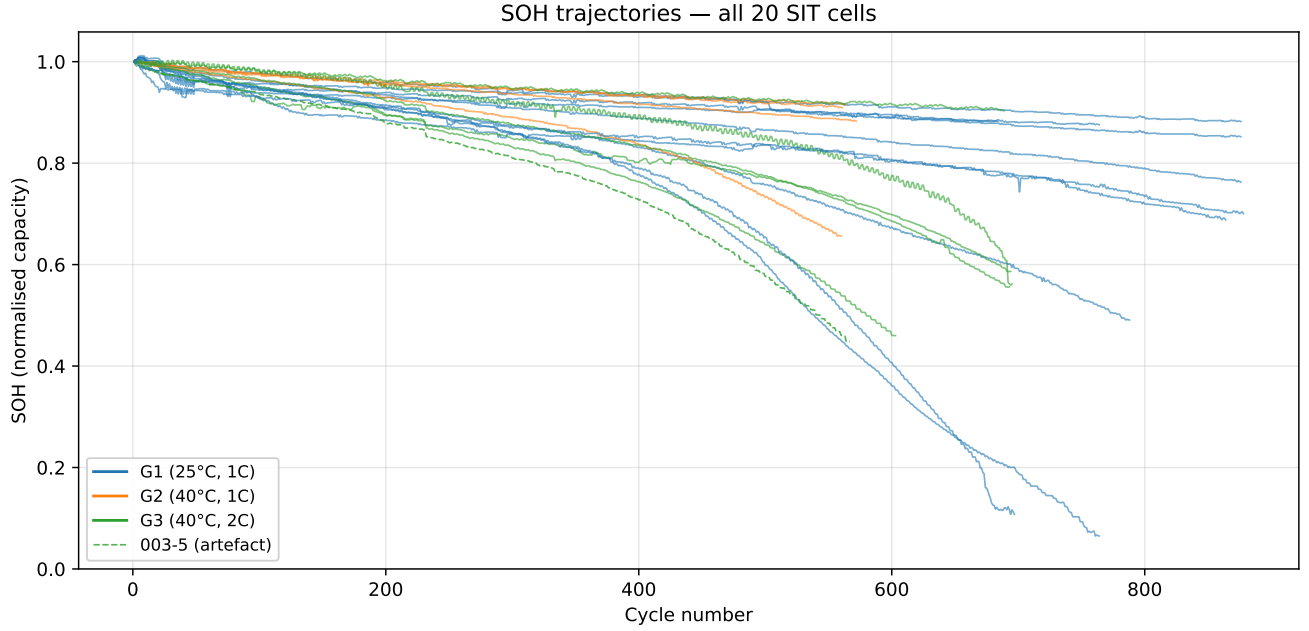


Figure 1: SOH trajectories for all 20 SIT cells.

Table 4

DE-LSTM baseline RMSE by operating-condition group.

Group	Condition	RMSE (30%)	RMSE (50%)	RMSE (70%)
G1	25 °C, 1C	0.030 ± 0.025	0.012 ± 0.005	0.010 ± 0.006
G2	40 °C, 1C	0.018 ± 0.021	0.011 ± 0.007	0.011 ± 0.008
G3 [†]	40 °C, 2C	0.020 ± 0.021	0.011 ± 0.008	0.010 ± 0.008
All [†]		0.025 ± 0.023	0.011 ± 0.006	0.010 ± 0.007

[†] Excluding the recovery-artefact cell 003-5, whose 30%-split RMSE (0.234) is dominated by its SOH>1 segment (50/70%: 0.017/0.014).

Table 5

Single-cell predictor and trivial-baseline RMSE on the SIT test cells.

Method	RMSE (30%)	RMSE (50%)	RMSE (70%)
<i>Recurrent (LSTM family)</i>			
Simple LSTM	0.0661	0.0352	0.0239
PA-LSTM [3]	0.0158	0.0263	0.0173
GA-LSTM [4]	0.0633	0.0198	0.0122
DE-LSTM [5]	0.0379	0.0118	0.0105
<i>Non-recurrent (modern)</i>			
TCN	0.1078	0.1017	0.1089
Transformer	0.1681	0.1130	0.1001
<i>Trivial baselines</i>			
Persistence ($\hat{y}_t = y_{t-1}$)	0.0123	0.0128	0.0134
Drift ($\hat{y}_t = 2y_{t-1} - y_{t-2}$)	0.0210	0.0219	0.0228
Linear AR	0.0068	0.0052	0.0052
Cubic curve fit	1.499	0.158	0.088

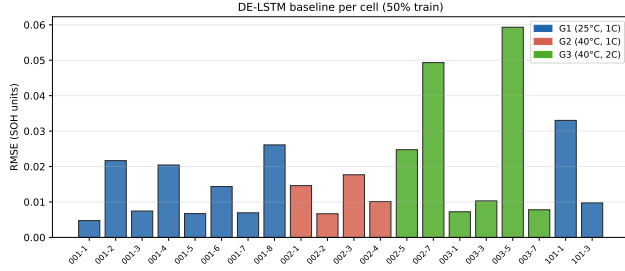


Figure 2: DE-LSTM per-cell RMSE at the 50% split for all 20 SIT cells, sorted by final SOH.

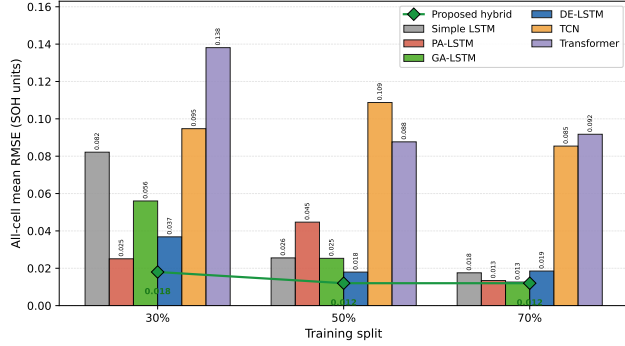


Figure 3: Mean RMSE of six single-cell predictors and the proposed hybrid across the 30%, 50% and 70% splits.

from *observed* windows is a near-persistence task on smooth SOH trajectories: successive cycles differ by a fraction of a percent, the predictor always receives the true recent history, and a linear map over the last 20 observations is essentially optimal. The task never requires anticipating a knee or a crash, only following one cycle behind it, which is also why the curve-fitting baseline, the one trivial method forced to extrapolate, fails by an order of magnitude. We term this **benchmark saturation**: the protocol under which this method lineage, and much of the field, ranks models cannot in fact distinguish them. A two-way variance decomposition over the six deep predictors makes the point quantitatively: the identity of the *cell* explains 58%/31%/25% of prediction-error variance at the 30%/50%/70% splits respectively, whereas the choice of *method* explains only 15% to 17% at every split, with the remainder attributable to cell-method interaction and training-run variance. Model selection should therefore rest not on one-step leaderboards but on capabilities the task actually demands: predicting cells with *no* usable history, quantifying uncertainty honestly, and transferring across operating regimes and chemistries, the capabilities evaluated in Section 7.

5.2. Cell-to-Cell Variability and the Two-Sided Failure Mode

We examine this cell-to-cell variability from three angles: how predictable each cell is on its own, what happens when a model is pooled across cells, and whether early-life data can identify the problem cells in advance.

5.2.1. Variability at Matched Exposure

The 13.2× spread in final SOH (0.067 to 0.885) compares cells at different points in their lives, because they reach end of life after different cycle counts. To measure the variability fairly we compare the G1 cells at matched exposure. At a common cycle count, the shortest G1 record of 684 cycles, SOH ranges from 0.20 to 0.90, a 7.6× ratio; at a common cumulative discharge throughput of 26.8 kWh it ranges from 0.07 to 0.92, a 13.7× ratio. The spread therefore remains roughly 8 to 14 times under every matched-exposure definition, so it is not an artefact of unequal test durations but a genuine dispersion among nominally identical cells.

5.2.2. Within-Cell Predictability

A cubic polynomial fit to each G1 cell's SOH trajectory achieves a mean R^2 of 0.964 (minimum: 0.908 for cell 001-6), confirming that individual trajectories are highly predictable given sufficient training data. On average 3.6% of the trajectory variation is not captured by this particular cubic fit; this residual reflects the choice of fit and is not an irreducible noise floor. The challenge is therefore not within-cell prediction but *between-cell diversity*: knowing a cell's past trajectory does not reveal where it sits in the manufacturing distribution.

5.2.3. Cross-Cell LOOCV: A Two-Sided Failure Mode

Leave-one-out cross-validation (LOOCV) on the 10 G1 cells reveals a nuanced picture (Table 6, whose factor column gives the cross-cell/same-cell RMSE ratio, with values below 1 favouring cross-cell). The pooled cross-cell model (a plain LSTM, $H = 64$, $W = 20$, trained on the *full* trajectories of the other nine cells and tested on the held-out cell) achieves a lower *mean* RMSE than same-cell training (which uses only the first 50% of the held-out cell; 0.049 vs. 0.060 at 50% split), but this average conceals opposing failure modes:

Table 6

Same-cell vs cross-cell RMSE for the 10 G1 cells (50% split).

Cell	SOH _{end}	Same-cell	Cross-cell	Factor
001-1	0.767	0.0363	0.0197	0.54×
001-2	0.878	0.0149	0.0262	1.77×
001-3	0.493	0.0938	0.0628	0.67×
001-4	0.709	0.0348	0.0389	1.12×
001-5	0.885	0.0166	0.0340	2.04×
001-6	0.854	0.0153	0.0189	1.24×
001-7	0.885	0.0162	0.0311	1.92×
001-8	0.067	0.1746	0.1183	0.68×
101-1	0.696	0.0424	0.0425	1.00×
101-3	0.118	0.1553	0.1015	0.65×
Mean		0.0600	0.0494	0.82×

The cross-cell model is better for 4 of 10 cells, and the largest gains are on the catastrophic degraders (001-8: same-cell RMSE = 0.175, cross-cell = 0.118; 101-3: 0.155 vs. 0.102), where same-cell training on the first 50% of a catastrophically failing cell provides a severely biased trajectory

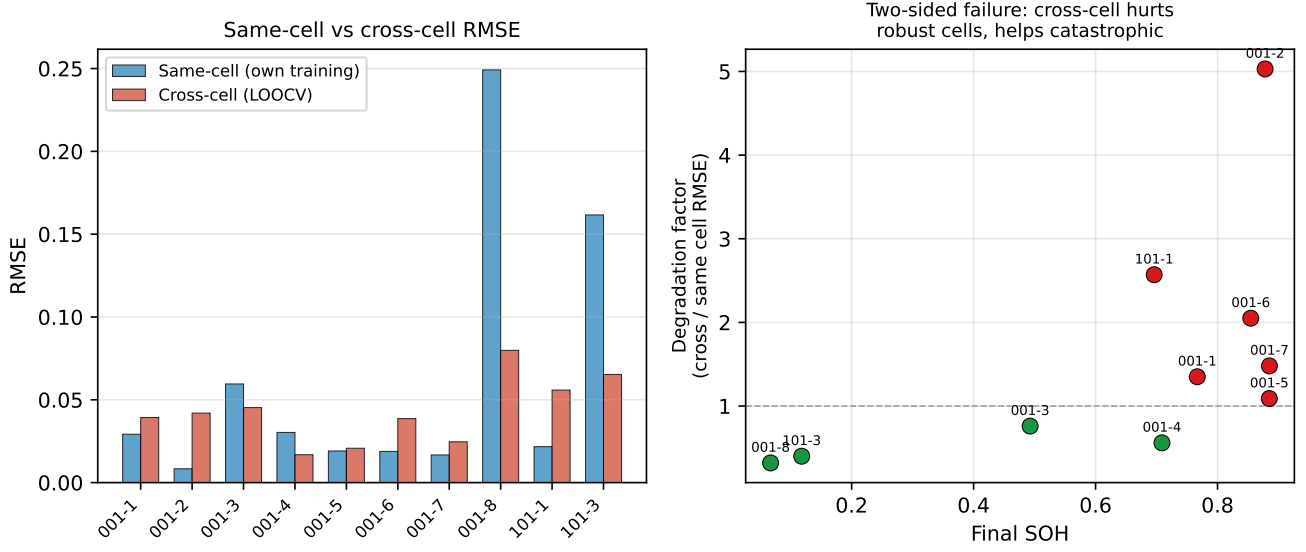


Figure 4: G1 LOOCV: same-cell vs cross-cell RMSE at the 50% split.

estimate. But pooling degrades five of the remaining cells by up to $2\times$ (001-5: 0.017 same-cell vs. 0.034 cross-cell; 001-2 and 001-7 are also harmed by factors of 1.8 to 1.9), and nothing in a cell’s early data reveals in advance whether it will be helped or harmed: the harmed cells are the *most robust* cells in the population, whose own smooth history is a better guide than the pooled average. The split is not sensitive to the training budget: when the cross-cell model is retrained on only the first 50% of each source cell (matching the same-cell information budget), the mean falls to 0.019 vs. 0.023 same-cell and the helped/harmed split remains 6/10, so the two-sided pattern is a property of pooling itself, not of the data advantage.

Figure 4 makes this pattern visual: the left panel contrasts same-cell and cross-cell RMSE for each G1 cell, while the right panel plots each cell’s degradation factor against its final SOH, showing cross-cell training helping the low-SOH catastrophic degraders but hurting the high-SOH robust cells. This **two-sided failure mode**, where cross-cell training rescues catastrophic degraders but unpredictably harms a minority of ordinary cells, motivates cell-specific adaptation via the fingerprint encoder in Section 7.1.

5.2.4. Early-Life Fingerprint

The practical value of this insight depends on whether the early SOH trajectory (observable in the first weeks of deployment) can identify which cells are likely catastrophic degraders. Table 7 reports Pearson correlations between early-life statistics and final SOH across 47 MIT LFP cells ($\text{SOH}_{\text{end}} < 1$, excluding 2 capacity-recovery artefacts).

The 50-cycle slope ($r = -0.38$, $p = 0.008$) and 10-cycle max_drop ($r = +0.37$, $p = 0.010$) are the most predictive individual features (Figure 5), but with $R^2 \approx 0.14$ these statistics explain only 14% of the variance in final SOH. Linear early-life statistics are insufficient for reliable individual-cell prediction; this motivates the CNN

Table 7

Correlation of early-life features with final SOH.

K cycles	Feature	r	p	Sig.
10	early_std	+0.322	0.027	*
10	max_drop	+0.371	0.010	*
10	slope	-0.073	0.625	
30	slope	-0.297	0.043	*
50	slope	-0.380	0.008	**

Significance: ** $p < 0.01$; * $p < 0.05$.

fingerprint encoder, which learns a nonlinear representation from the full 30-cycle trajectory rather than hand-crafted summary statistics.

5.3. Cross-Condition Transfer Is Dominated by Training-Run Variance

Table 8

Cross-condition transfer RMSE between operating-condition groups.

Train	Test	RMSE	Factor vs G1 LOOCV
G1 (LOOCV)	G1	0.062 ± 0.065	1.0 $\times$ (baseline)
G1	G2	0.008 ± 0.002	0.1 $\times$
G1	G3	0.034 ± 0.041	0.6 $\times$
G2	G3	0.145 ± 0.139	2.4 $\times$
G1+G2	G3	0.145 ± 0.133	2.4 $\times$

Table 8 reports the cross-condition transfer RMSE (mean $\pm$ std across the target cells), with the degradation factor giving each transfer’s RMSE relative to the within-distribution G1 LOOCV baseline; Figure 6 plots the same values, with the dashed line marking that baseline. Two confounds emerge, and together they carry a methodological warning. First, G1→G2 (temperature shift only) yields a far lower RMSE

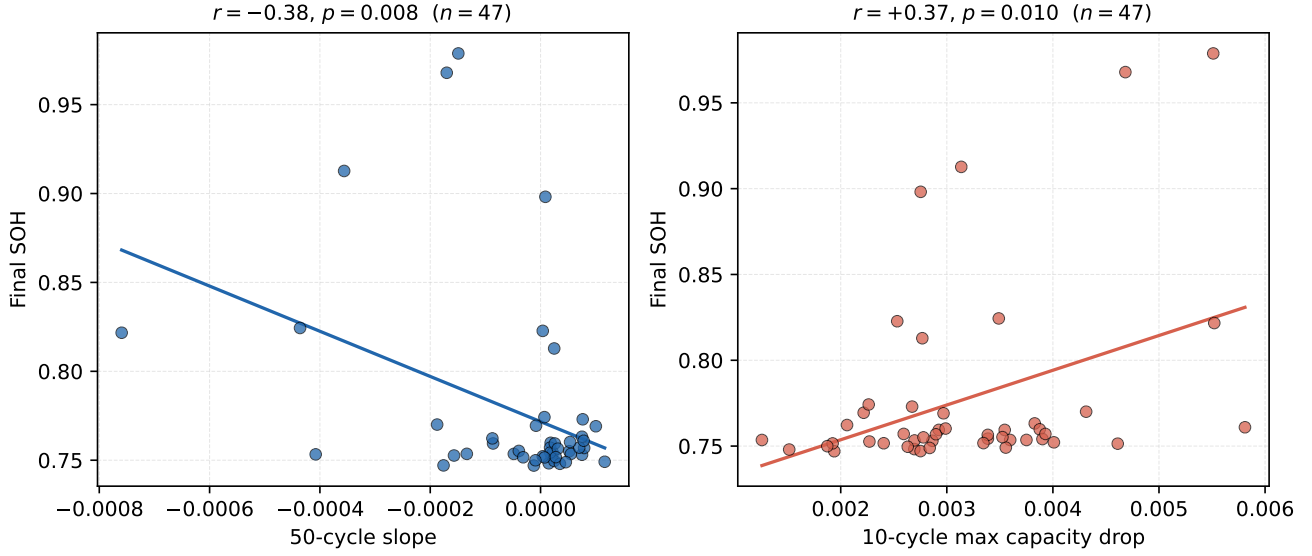


Figure 5: Early-life predictors of final SOH: 50-cycle slope and 10-cycle maximum capacity drop.

(0.008) than the G1 LOOCV baseline (0.062), not because temperature transfer is easy, but because G2 cells happen to have smooth, high SOH trajectories well within the G1 training distribution: cross-condition RMSE comparisons are confounded by the difficulty distribution of the test set. Second, and more consequentially, the transfer factors are dominated by *training-run variance* rather than by the condition shift itself. Each transfer row is one pooled model, and independent training draws of the same experiment produce qualitatively different conclusions: an earlier run of G1→G3 with library-default (unseeded) initialisation yielded 2.3× degradation, whereas the deterministic pipeline reported here yields 0.6×, i.e. *better* than within-distribution. Even within Table 8 the instability is visible: training on G1 alone transfers to G3 at 0.034, yet *adding* G2 data (G1+G2→G3) worsens the same transfer to 0.145, a difference no condition-shift mechanism explains. At one-step granularity, cross-condition effects on these data are smaller than the noise floor of neural-network training itself, another face of the benchmark saturation established in Section 5.1. We therefore do not claim a cross-condition brittleness effect; the defensible claims are the test-set difficulty confound and the dominance of run variance, both of which argue for capability-based evaluation with uncertainty quantification rather than single-model RMSE comparisons.

The within-distribution baseline used here (G1 LOOCV, 0.062) trains on the first 50% of each source-group cell, identical to the cross-condition transfers, so every degradation factor in Table 8 is computed within one consistent protocol. It is therefore larger than the full-trajectory cross-cell LOOCV reported in Section 5.2 (0.049), which trains on the complete trajectories of the other nine cells and so sees their degradation tails; the two baselines answer different questions and should not be compared directly.

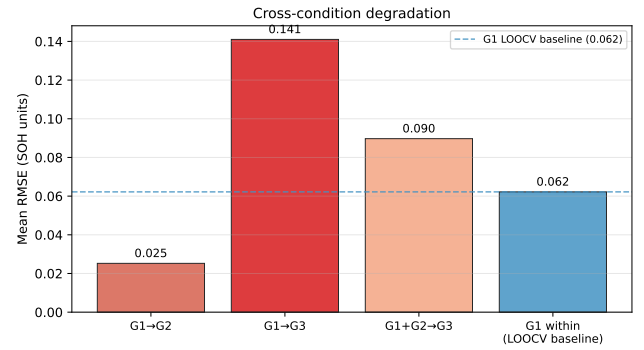


Figure 6: Mean RMSE for each cross-condition transfer under the deterministic pipeline.

5.4. Requirements for a Robust Predictor

The preceding results expose why no single-cell predictor stands out on SIT and what a better model must instead provide. The limiting factor is not the sequence model or its hyperparameters but cell-to-cell variability. Under nominally identical G1 conditions, per-cell optimised DE-LSTM RMSE still varies $\sim 5\times$ across the ten cells (Section 5.1), because a model trained on the early portion of a single cell's trajectory cannot tell where that cell sits in the manufacturing distribution: the first half of a catastrophic degrader closely resembles that of a moderate one. No amount of per-cell hyperparameter search resolves this, because the missing information is not present in the individual cell's early history.

Pooling data across cells, the obvious alternative, introduces a two-sided failure mode (Section 5.2). A single model trained on the whole population helps catastrophic degraders, for which the population average is a better prior than their own ambiguous early trajectory, but it degrades half of the ordinary cells by up to $2\times$, and which cells suffer

is not predictable a priori. Neither per-cell tuning nor naive pooling addresses the underlying problem.

These observations, together with the saturation of the one-step metric, reframe the task. What matters is not squeezing the last fraction of a percent from a saturated leaderboard but providing capabilities the leaderboard does not measure: a degradation prior broad enough to recognise extreme trajectories, prediction for cells with little or no usable history, and honest uncertainty estimates. The hybrid model provides the first through pretraining on a large, diverse LFP corpus (Section 4.1), the second through an early-life fingerprint encoder that enables prediction without target-cell fine-tuning, and the third through MC-Dropout with conformal calibration.

6. Proposed Method: Hybrid Model

This section presents the proposed hybrid model: an overview of its three phases, the architecture and training of each, the mechanism that distinguishes the fingerprint encoder from conventional fine tuning, and the design of the ablation used to isolate each component.

6.1. Framework Overview

Figure 7 summarises the proposed method in two steps. Step 1 (top) builds the hybrid in three phases: it pretrains on past LFP cells, then, for each new cell, forms an early-life fingerprint and fine tunes. Step 2 (bottom) shows the three outputs this supports, one per deployment question (how healthy the cell is now, which cells will fail, and how long each cell will last), and pairs each output with the experiment that validates it. Every battery dataset fed into the model is marked with a battery icon. The core is the three-phase hybrid that predicts next-cycle SOH, detailed below and evaluated in Section 7. Phase 1 pretrains the hybrid network, the convolutional fingerprint encoder together with the LSTM, end-to-end on the chemistry matched MIT dataset to acquire general LFP degradation knowledge. Phase 2 applies the pretrained encoder to the first 30 discharge cycles of a target cell, producing a fingerprint that initialises the LSTM hidden state and conditions all subsequent predictions. Phase 3 fine tunes the LSTM on the target cell's training split and performs MC-Dropout inference to deliver uncertainty-aware predictions whose prediction intervals are empirically validated rather than merely assumed.

The same early-life representation supports two capabilities developed later in the paper. An early screen for manufacturing-defective cells reads the discharge voltage curve rather than the capacity trajectory (Section 8), and a remaining-useful-life forecaster replaces the one-step head with a direct multi-horizon decoder (Section 9). The two transfer levers are distinct: the base hybrid is pretrained on the chemistry-aligned MIT LFP set, whereas the RUL decoder is pretrained on a 155-cell LFP foundation corpus drawn from the MIT, HUST, and SNL collections. These two levers are matched to two different tasks rather than merged into one: one-step SOH is a local prediction whose

accuracy saturates on these smooth large-format trajectories (Section 7), so the chemistry-aligned MIT corpus is sufficient and additional data brings no measurable gain, whereas remaining-useful-life forecasting is a long-horizon, data-limited task whose error falls only when the corpus is enlarged to the full 155-cell collection (Section 9.2). Because that foundation set contains the MIT cells used for the base hybrid, the two corpora are nested rather than independent, so the RUL stage extends the SOH stage rather than competing with it; we keep the two pretraining stages separate only so that each transfer lever can be isolated in the ablation.

6.2. Hybrid Model Architecture

The hybrid model operationalises the two capabilities identified in Section 5.4 through three phases, drawn as the Step 1 sequence at the top of Figure 7:

6.2.1. Phase 1: MIT Pretraining

The complete hybrid architecture (the convolutional fingerprint encoder of Phase 2 together with the LSTM) is pretrained *end-to-end* on the 40-cell MIT LFP corpus (Section 4.1). For each cell, the first 30 cycles form the fingerprint input and the network is trained to predict every *subsequent* cycle's SOH from a recent look-back window, minimising next-cycle mean squared error (LSTM: $H = 64$, $W = 20$; Adam, learning rate 10^{-3} ; early stopping on a 10% validation split). This single objective jointly learns two things: general LFP degradation dynamics in the LSTM, and the mapping from early-life trajectory shape to the LSTM initial state in the fingerprint encoder. The encoder weights obtained here are reused, and frozen, in Phase 3.

6.2.2. Phase 2: Convolutional Fingerprint Encoder

A 1-D convolutional network (two Conv1D layers with 32 and 16 filters respectively, kernel size 3, followed by global average pooling and a dense projection) encodes the first 30 SOH cycles of the target cell into a 16-dimensional embedding vector. This encoder is the same one trained jointly in Phase 1; its parameters are not re-learned per target cell. Two dense layers project this embedding to the LSTM initial hidden state h_0 and cell state c_0 , conditioning predictions on the individual cell's observed early behaviour.

Formally, let $\mathbf{s} = [s_1, s_2, \dots, s_{30}]^T \in \mathbb{R}^{30}$ denote the first 30 normalised discharge capacities of the target cell. The encoder computes:

$$\mathbf{z} = \text{GlobalAvgPool}(\text{Conv}_2(\text{Conv}_1(\mathbf{s}))) \in \mathbb{R}^{16} \quad (2)$$

$$h_0 = \tanh(\mathbf{W}_h \mathbf{z} + \mathbf{b}_h), \quad c_0 = \tanh(\mathbf{W}_c \mathbf{z} + \mathbf{b}_c) \quad (3)$$

where $\mathbf{z}$ is the fingerprint embedding from Equation (2), $\mathbf{W}_h$ and $\mathbf{W}_c$ are the weight matrices of the two dense projection layers, $\mathbf{b}_h$ and $\mathbf{b}_c$ are their bias vectors, and h_0 and c_0 are the resulting LSTM initial hidden and cell states. The critical point is that $\mathbf{s}$ is treated as a *static shape*, not a sequence to continue: the CNN pools over all 30 cycles simultaneously

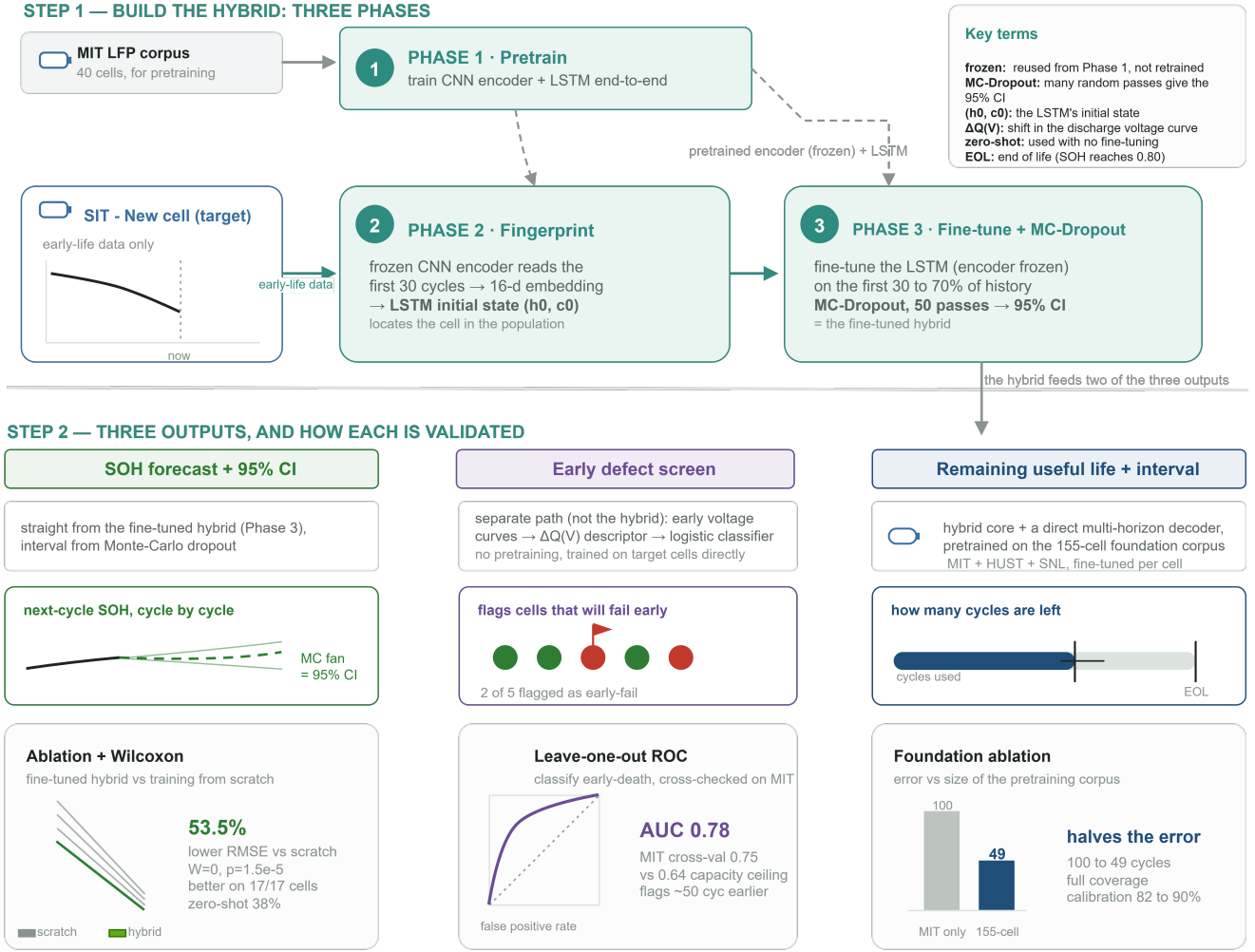


Figure 7: The proposed method: the three-phase hybrid and its three validated outputs.

to extract curvature, rate of decline, and early-knee features that characterise where this cell sits in the manufacturing distribution relative to the pretrained population.

6.2.3. Phase 3: Fine tuning and MC-Dropout

The pretrained LSTM is fine tuned on the first $train_pct\%$ of the target cell's SOH trajectory, starting from the cell-specific initial state (h_0, c_0) produced by the fingerprint encoder. The fingerprint encoder weights, learned in Phase 1, are held *frozen* throughout fine tuning; only the LSTM and output-layer weights are updated. Training and inference use fixed-length look-back windows of $W = 20$ cycles drawn from the target cell's SOH. Each window is processed independently: the LSTM is re-initialised to the fingerprint-derived state (h_0, c_0) at the first step of every window rather than carrying hidden state across windows, so each prediction is conditioned on both the cell-level fingerprint and its local W -cycle history. The fine tuning objective is:

$$\mathcal{L}_{FT} = \frac{1}{K} \sum_{i=1}^K (\hat{y}_i - y_i)^2, \quad K = \left\lfloor \frac{p}{100} \cdot T \right\rfloor \quad (4)$$

where y_t is the observed SOH at cycle t , $\hat{y}_t$ is the model prediction, T is the total cycle count of the target cell, and $p \in \{30, 50, 70\}$. Dropout (rate 0.2) applied to the LSTM output during both training and inference provides uncertainty estimates via 50 stochastic forward passes. The 95% prediction interval is $\hat{y} \pm 1.96 \hat{\sigma}$ (an approximate MC-Dropout interval, not a calibrated prediction interval), where $\hat{\sigma}$ is the empirical standard deviation across the 50 passes.

6.3. Pretraining-Run Variance and Checkpoint Selection

Pretraining a neural network is stochastic: independent runs on the identical corpus, differing only in random initialisation, converge to representations of very different transfer quality. Across five deterministic pretraining runs of Phase 1, downstream SIT improvement over scratch ranged from -32% to $+52\%$: three runs learned genuine degradation dynamics while two collapsed toward a persistence solution that transfers poorly. Crucially, *source-domain criteria failed to identify the collapsed runs*: MIT validation loss was essentially identical for good and collapsed runs ($\approx 3.4 \times 10^{-3}$), one-step-ahead RMSE on held-out MIT cells was

Table 9

Comparison of the two mechanisms that use early-cycle SOH data.

	Incremental fine tuning	Fingerprint encoder
<i>Input view</i>	Time series $s_1 \rightarrow s_2 \rightarrow \dots \rightarrow s_K$	Static shape $s \in \mathbb{R}^{30}$
<i>Operation</i>	Adjust LSTM weights θ to minimise $\mathcal{L}_{FT}$	Map shape to (h_0, c_0) via CNN
<i>Output</i>	Updated model weights	Cell-specific initial state
<i>Question answered</i>	What comes next in this sequence?	What is this cell's early-life degradation signature?
<i>Limitation</i>	Ambiguous for extreme cells in the first $p\%$	Requires a diverse pretraining corpus

inverted (collapsed runs win it, because persistence excels at one-step prediction), and recursive multi-step rollout was uninformative.

We therefore adopt a simple target-domain selection protocol:

1. pretrain $k = 5$ candidate checkpoints with independent seeds;
2. evaluate each candidate's fine-tuned RMSE on the three designated *calibration cells* (Section 4.3);
3. deploy the checkpoint with the lowest mean calibration RMSE.

The calibration cells are chosen by an a-priori rule (the numerically first cell of each condition group) and are excluded from all reported results. The protocol is robust to the choice of calibration cells: exhaustively enumerating all $\binom{20}{3} = 1140$ possible three-cell calibration sets, every one selects a checkpoint with positive test improvement (100%). In deployment terms the protocol is realistic: a manufacturer qualifying a battery product retains laboratory cells of that product, and three such cells suffice. All results in Section 7 use the selected checkpoint; the full five-run sensitivity study is released with the code.

6.4. Mechanism Contrast: Fingerprint Encoder vs. Fine Tuning

Although Phases 2 and 3 both draw on the same early-cycle SOH data, they use it in fundamentally different ways, a distinction that directly motivates the ablation study in Section 6.5. Table 9 makes this contrast explicit.

For cells at the extremes of the manufacturing distribution (catastrophically fast: cell 001-8; exceptionally robust: cells 001-5 and 001-7), the first 50% of the SOH trajectory is *ambiguous*: it resembles a moderate degrader. Fine tuning on this ambiguous first half fits a moderate trajectory and projects it forward, substantially underestimating the second half. The fingerprint encoder, by contrast, conditions the LSTM on the early-life trajectory shape (curvature and rate of decline across the first 30 cycles), placing the prediction in the context of the pretrained population. Whether this conditioning encodes precise cell *identity* or a more generic in-distribution prior is an empirical question; the fingerprint control study in Section 7.1.2 settles it clearly in favour of the latter, as zero, mean, shuffled and random fingerprints all match the correct one, and we interpret the mechanism as population conditioning accordingly.

6.5. Ablation Study Design

To isolate the contribution of each hybrid component, four configurations are compared:

1. **Scratch LSTM**: Train the hybrid's LSTM backbone ($H=64$, $W=20$) from scratch on the target cell's training split only, with no pretraining, no fingerprint, and no differential-evolution hyperparameter search. This is the ablation control for the hybrid and is distinct from the DE-optimised DE-LSTM baseline of Section 5.1; the two therefore differ in RMSE.
2. **Fingerprint-conditioned transfer (pretrain + fingerprint, no fine tuning)**: the MIT pretrained model with the fingerprint-derived initial state applied directly to the target cell, with *no fine tuning* on target data (the only target-cell information used is the first 30 cycles, from which the fingerprint is computed; no target SOH label is fitted). Isolates the value of pretraining and the fingerprint before any target-cell adaptation.
3. **Pretrain + fine tune (no fingerprint)**: MIT pretrained model fine tuned on the target cell's training split, but without the fingerprint encoder. Quantifies the marginal value of the fingerprint over fine tuning alone.
4. **Full hybrid**: All three phases (pretrain + fingerprint + fine-tune).

6.6. Reproducibility and Determinism

All experiments run under a fully deterministic configuration: a single global seed set via `tf.keras.utils.set_random_seed` (which, unlike `tf.random.set_seed` alone, also seeds Keras weight initialisers), TensorFlow op-determinism enabled, and float32 precision throughout. Under this configuration every number reported in this paper reproduces byte-for-byte from a fresh clone of the released code. The repository includes a single-command verification script (`verify_determinism.sh`) that exercises each experiment family's actual training and inference code path twice in separate processes and byte-compares the outputs, so that reviewers and users can confirm determinism on their own hardware before relying on the pipeline. We adopted this discipline after observing that library-default (unseeded) training produced run-to-run variation large enough to alter experimental conclusions (Sections 5.1 and 5.3); we regard it as a necessary condition for benchmark claims in this field.

7. Hybrid Model Evaluation

This section evaluates the hybrid through an ablation isolating each component, the main accuracy results on the 17 SIT test cells, conformal calibration of its uncertainty intervals, and a cross-dataset validation of the chemistry-alignment effect.

7.1. Ablation Study and Main Results

We first isolate the contribution of each component, then report the full hybrid's accuracy and calibration across the 17 test cells (the three calibration cells of Section 4.3 are excluded from every reported metric).

7.1.1. Ablation Study

Table 10

Ablation study: contribution of each hybrid component (50% split).

Configuration	RMSE	Improvement vs scratch	PI Coverage
Scratch LSTM	0.043 ± 0.050	(baseline)	n/a
Fingerprint transfer (MIT + FP, no FT)	0.030 ± 0.031	23.3% [‡]	74.3%
Pretrain + FT (no fing.)	0.018 ± 0.017	44.4% [‡]	79.0%
Full hybrid (proposed)	0.016 ± 0.014	51.1%[‡]	80.8%

[‡] Wilcoxon $p < 0.001$ vs scratch.

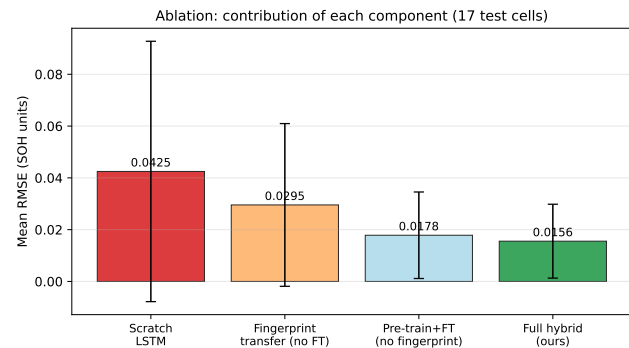


Figure 8: Ablation study: mean RMSE per configuration (50% split).

Table 10 and Figure 8, which plots the same mean RMSE per configuration, show that every component contributes and the effects compose monotonically: fingerprint-conditioned transfer (pretrained model with the fingerprint-derived initial state, *no* fine tuning on target data) already achieves a 23.3% improvement over scratch and beats scratch on 13 of 17 test cells; adding fine tuning without the fingerprint reaches 44.4%; and the full hybrid reaches 51.1% (all Wilcoxon $p < 0.001$). Three observations stand out. First, fine tuning is the larger and more dependable lever: it accounts for the bulk of the improvement, whereas fingerprint-conditioned transfer recovers under half of the full hybrid's benefit before any adaptation is possible. That transfer capability is nonetheless one no other evaluated method offers, giving RMSE 0.030 for a cell with *no* target-cell training, from 30 observed cycles. Second, the fingerprint and fine tuning are complementary rather than redundant: the fingerprint on top of pretraining+fine tuning still reduces RMSE

from 0.018 to 0.016. Third, the fingerprint conditioning works chiefly by placing the LSTM in the *pretrained population's* operating regime (any in-distribution early-life shape suffices) rather than by encoding a precise cell identity. This reading is reinforced by the fingerprint-control analysis of Section 6.4, where a shuffled or population-mean fingerprint performs close to the correct one, and by the finding that the trained encoder maps all cells to a nearly constant embedding: the mechanism is a population prior, not cell-specific identification. Prediction-interval coverage improves in the same monotone order (74.3% → 79.0% → 80.8%), so with a well-selected pretrained checkpoint there is no accuracy versus calibration trade-off among the configurations.

7.1.2. What the fingerprint actually contributes

To test whether the fingerprint encodes cell-specific information or merely a population prior, we fine-tune and predict with five fingerprint variants: the correct cell fingerprint, a zero vector, the population-mean fingerprint, a shuffled fingerprint (another cell's), and a random fingerprint. Table 11 shows all five give indistinguishable accuracy (RMSE 0.0149 to 0.0156) and coverage (80 to 82%); the correct fingerprint is, if anything, marginally worse than the controls. A direct check of the encoder confirms the reason: across the MIT cells the embeddings have a mean pairwise cosine similarity of essentially 1.0, so the trained encoder maps every cell to a near-constant vector. The fingerprint therefore carries no cell-specific information; it acts as an in-distribution population prior, placing the LSTM in the pretrained regime, rather than identifying the individual cell. We report this openly and, throughout, describe the mechanism as population conditioning rather than cell-specific fingerprinting.

Table 11

Fingerprint control study (50% split, 17 test cells).

Fingerprint	RMSE	PI coverage
Correct (this cell)	0.0156	80.2%
Population mean	0.0149	81.3%
Shuffled (another cell)	0.0149	81.4%
Random	0.0149	81.4%
Zero	0.0149	81.5%

7.1.3. Hybrid Accuracy Across Operating Conditions

Table 12

Full hybrid results by operating-condition group (50% split).

Group	n	Hybrid RMSE	Scratch RMSE	Improvement	PI Coverage
G1 (25°C, 1C)	9	0.015 ± 0.011	0.035 ± 0.034	42.3%	77.8%
G2 (40°C, 1C)	3	0.006 ± 0.002	0.016 ± 0.013	54.4%	75.5%
G3 (40°C, 2C)	5	0.023 ± 0.021	0.073 ± 0.077	65.1%	89.6%
All SIT (test)	17	0.016 ± 0.014	0.043 ± 0.050	51.1%	80.8%

The full hybrid model achieves a statistically significant improvement over scratch LSTM across the 17 test cells (Wilcoxon $W = 0$, $p = 1.5 \times 10^{-5}$; hybrid better in 17/17

cells). Table 12 breaks the hybrid and scratch RMSE, the mean improvement, and the PI coverage down by operating-condition group. Figure 9 shows the per-cell results: the top panel compares each cell's scratch and hybrid RMSE, while the bottom panel gives the corresponding per-cell improvement of the hybrid over scratch (positive for every test cell; the smallest improvement is 15.2% on cell 101-1).

On the one-step metric itself, every method is bunched on a saturated scale: the hybrid's mean RMSE is 0.0241, 0.0156 and 0.0148 at the 30%, 50% and 70% splits (Figure 3), the per-cell-tuned DE-LSTM reaches 0.0118 and 0.0105 at 50% and 70%, and even a trivial linear autoregressor sits near 0.005, differences that carry little information on this saturated benchmark (Section 5.1). The hybrid's decisive advantage is therefore not a marginal one-step number but a set of capabilities that no other method on the leaderboard provides. It is the only method that operates with no target training at all (fingerprint-conditioned transfer, 0.030), the only method that quantifies uncertainty (Section 7.1.4), and the only one validated across three manufacturers (MIT, SIT and LISHEN), two chemistries (LFP and LCO) and three operating conditions (Section 7.2), whereas PA-LSTM, GA-LSTM and DE-LSTM provide point estimates only, for cells that already have an established history. On the cells that matter most for deployment, the catastrophic degraders, this advantage is also an accuracy advantage, as the next paragraph shows.

Within G1, the hybrid is particularly effective for catastrophic degraders (Figure 10): cell 001-8 ($\text{SOH}_{\text{end}} = 0.067$) improves 61.8% (0.038 vs. 0.098 scratch); cell 101-3 ($\text{SOH}_{\text{end}} = 0.118$) improves 78.5% (0.018 vs. 0.083). Both gains are driven by the MIT pretraining, which has encountered comparable extreme-degradation trajectories in the training population, a trajectory the scratch model cannot learn from 50% of a single catastrophic cell's history. Figure 10 plots the hybrid's predicted versus actual SOH with 95% MC-Dropout uncertainty bands on the held-out second half (50% split) for three representative cells, a robust G1 cell, the catastrophic degrader 001-8, and a condition-shifted G3 cell, with the training portion shaded and the train/test boundary dashed. Across all three the hybrid tracks both gentle and catastrophic trajectories, and the prediction interval widens where degradation is steepest.

For G3 test cells (40/2C), the hybrid achieves mean RMSE of 0.023, a 65.1% improvement over the scratch LSTM given the same 50% G3 fine tuning data (0.073; Table 12), by combining MIT-learned generic LFP priors with that fine tuning. Cold cross-condition transfer (a G1-trained model applied to G3 with no G3 data) is not a stable reference point, since Section 5.3 showed such estimates are dominated by training-run variance; the comparison against the equally-fine-tuned scratch model above is the fair like-for-like measure of the hybrid's contribution. The G3 conclusions are not an artefact of cell 003-5 (the $\text{SOH} > 1$ capacity-recovery cell): excluding it, the four remaining G3 test cells give a hybrid RMSE of 0.015 (scratch 0.039), a 63.1% mean per-cell improvement versus 65.1% with 003-5

included, so the artefact slightly inflates but does not drive the result.

7.1.4. Conformal Calibration of the Uncertainty Intervals

The raw MC-Dropout intervals are approximate (Section 6.2) and under-cover in practice: pooled over the 17 test cells, the empirical coverage of nominal 50/80/90/95/99% intervals is 47/74/79/82/88% (Figure 11, left). We recalibrate with conformal prediction [28], and two design choices matter. First, the cells used to select the pretraining checkpoint must not also calibrate the intervals, so we exclude the three selection cells from calibration. Second, hundreds of cycles from one cell are not independent samples: the residuals within a cell are strongly correlated, so we treat the cell, not the cycle, as the exchangeable unit and use grouped, leave-one-cell-out conformal, taking one normalised nonconformity score $|y_t - \hat{y}_t|/\hat{\sigma}_t$ quantile per cell and calibrating each test cell against the others. Under this honest protocol the intervals attain at least their nominal coverage, reaching 91% at nominal 80% and 100% at nominal 95%. The over-coverage is expected: only sixteen cells calibrate each quantile, so the finite-sample conformal correction is conservative and the intervals are correspondingly wide. For reference, the naive alternative that calibrates on the three selection cells reports a tighter 98% at nominal 95%, but it reuses the selection data and its apparent tightness is not trustworthy. Coverage is weakest on the two structurally atypical cells, the catastrophic degrader 001-8 and the recovery artefact 003-5, whose error distributions differ from anything in the calibration set, a reminder that conformal guarantees assume exchangeability, which extreme outliers violate. The full hybrid is the only method in Table 5 that produces uncertainty intervals at all; recalibrated this way they are conservative but honest for BMS threshold logic, while flagged outliers still require special handling.

7.1.5. Uncertainty Calibration: The Chemistry-Alignment Effect

Figure 12 plots the empirical 95% PI coverage per cell (left) and averaged per dataset group (right), and reveals a striking pattern: PI coverage on SIT LFP cells (80.8% mean) is nearly double that on NASA LCO cells (43.6%), despite using the same MC-Dropout rate (0.2) and the same nominal 95% target. The most prominent structural difference is chemistry: the MIT pretraining data are LFP cells (matched to SIT), whereas NASA cells are LCO (LiCoO₂ [6]; mismatched). The datasets also differ in cell format, laboratory and cycling protocol, so chemistry is the most likely, but not the only possible, contributor to the coverage gap; the cross-dataset study below helps disentangle it.

We therefore call this a **chemistry and domain alignment effect** rather than a principle, since chemistry is the most likely contributor but is not isolated from format, laboratory and protocol. It has a practical implication for BMS deployment: transfer learning preserves uncertainty calibration only when the pretraining and deployment chemistries

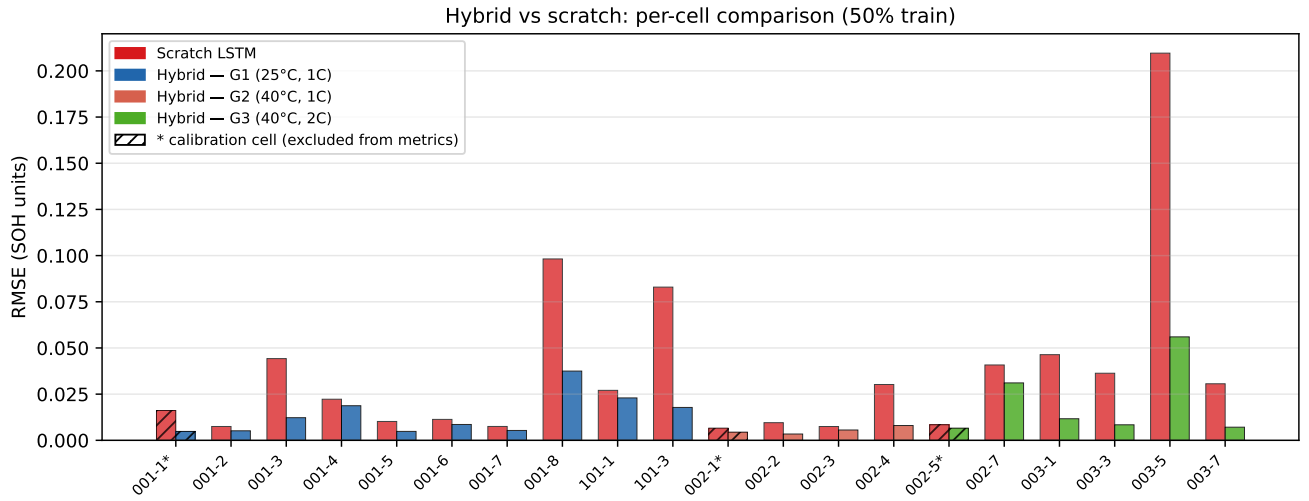


Figure 9: Hybrid vs. scratch LSTM on the SIT cells (50% split).

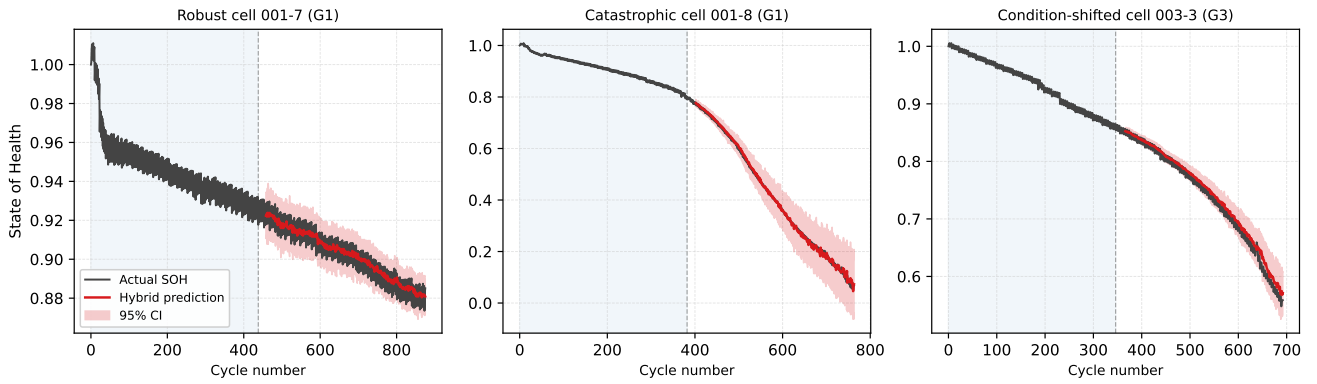


Figure 10: Hybrid predicted versus actual SOH with 95% MC-Dropout uncertainty bands (50% split).

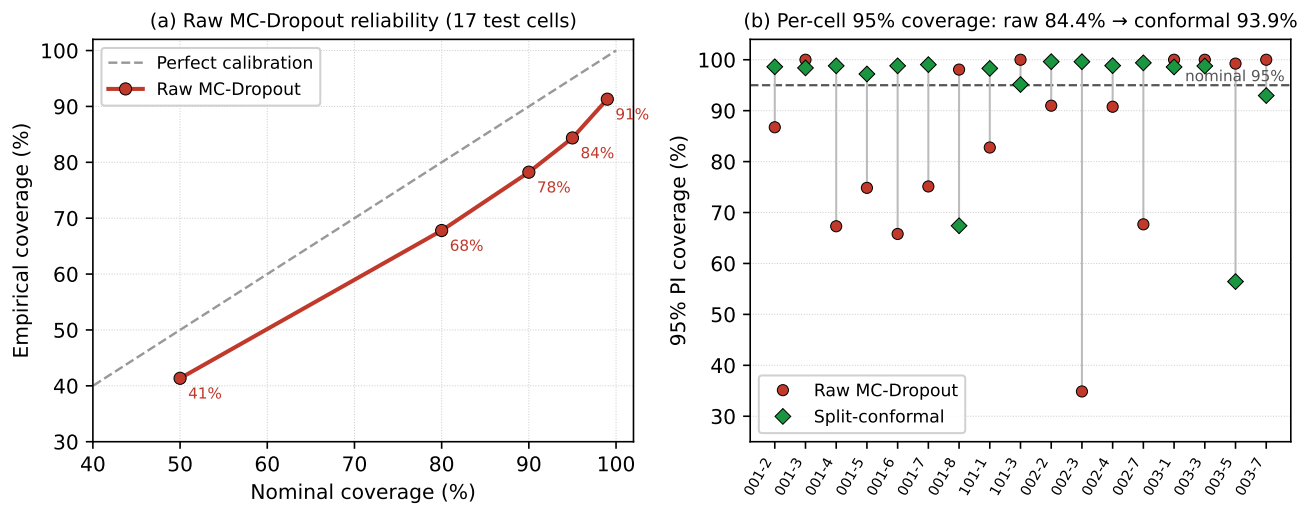


Figure 11: Uncertainty calibration on the 17 test cells.

Chemistry alignment: 17 SIT LFP test cells vs 26 NASA LCO cells

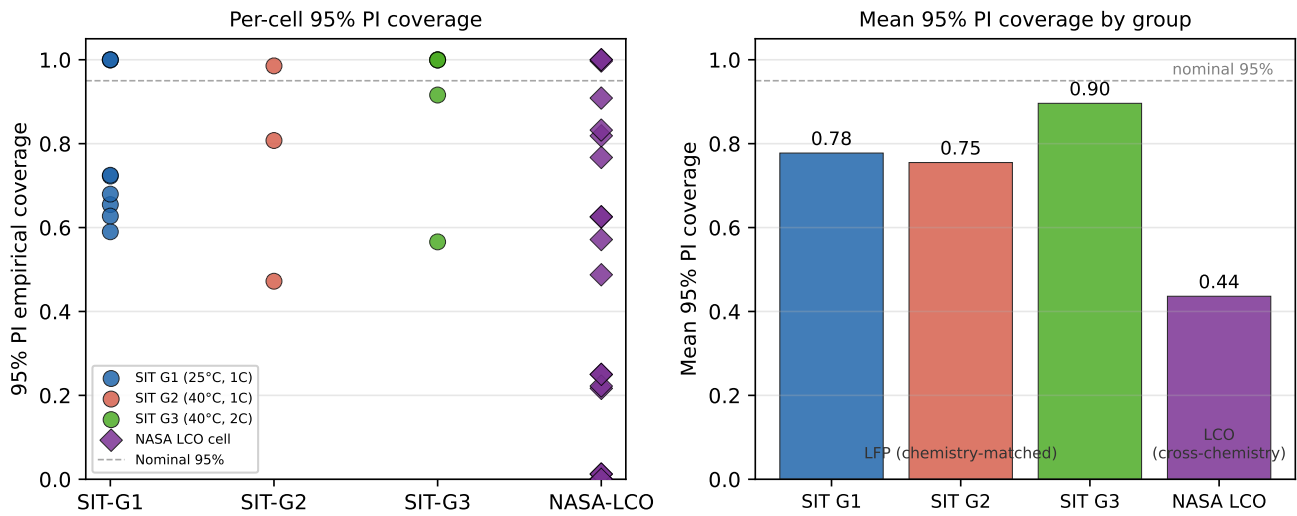


Figure 12: MC-Dropout 95% PI empirical coverage by dataset and group.

are matched. Deploying a model pretrained on LFP data to LCO cells not only degrades point estimate accuracy but also destroys uncertainty calibration, producing overconfident predictions that could mask true risk. Post-hoc calibration (conformal prediction [28] or isotonic regression) is required when chemistry is mismatched.

7.2. Cross-Dataset Validation of the Chemistry-Alignment Effect

The analysis in Section 7.1 established the chemistry-alignment effect from a single source-target pair (MIT→SIT). To verify that this finding generalises, we evaluate the hybrid framework across ten pairwise transfer scenarios spanning three LFP datasets (MIT Severson [18], SIT, and LISHEN 40 Ah LFP) and one LCO dataset (NASA 18650). The LISHEN dataset [29] provides three industrial-scale LFP cells (40 Ah) cycled at 2C with 20% depth of discharge (DOD) regular cycles and full-discharge reference tests every tenth cycle, offering a third independent LFP pretraining source distinct from both MIT and SIT in cell format, capacity, and cycling protocol.

Scenario A (MIT→SIT) repeats the main ablation of Table 10, included here for reference. In Table 13, the ten scenarios are grouped by chemistry alignment (LFP↔LFP matched, LCO↔LFP mismatched) and negative values denote degradation relative to scratch. Together with Figure 13, the results support three findings.

7.2.1. Fine tuning is the dependable adaptation mechanism

Pretraining plus fine tuning improves over scratch in *all ten* scenarios (+4.4% to +71.2%, mean 36.0%), and the full hybrid tracks it closely (mean 35.8%), being best or within one point of best in eight of ten (A, B, D, E, F, G, M and N). Whatever the source corpus, chemistry, or cell

format, fine tuning a pretrained model on the target cell's history is never worse than training from scratch in these experiments, which makes it the mechanism a practitioner can rely on when facing an unfamiliar dataset pair. The weakest transfers (M and N, both below 6%) share one cause: the 3-cell LISHEN corpus is too narrow a pretraining source for either mechanism.

7.2.2. Fingerprint-conditioned transfer is a distinct capability, not a universal accuracy win

Fingerprint-conditioned transfer performance varies far more across scenarios (−30.4% to +80.7%) than fine tuning does. It is strongly positive precisely where the pre-trained population resembles the deployment population: the paper's primary scenario A (MIT→SIT, +23.3%) and scenario G (MIT→LISHEN, +80.7%), both large LFP corpora transferring to LFP targets. It fails when the source poorly spans the target (H and M, both negative: SIT's 20 heterogeneous cells and LISHEN's 3 cells do not cover each other's regimes), because the fingerprint places a cell within the *pretraining* population's degradation distribution, and that placement misleads when the distribution does not contain the target's behaviour. The correct reading is therefore about regime, not rank: fingerprint-conditioned transfer is the only option during the first weeks of deployment, before any fine-tuning history exists, and it is trustworthy exactly when the practitioner can match pretraining chemistry and corpus breadth to the deployment population, precisely the condition a battery manufacturer qualifying one product line can guarantee.

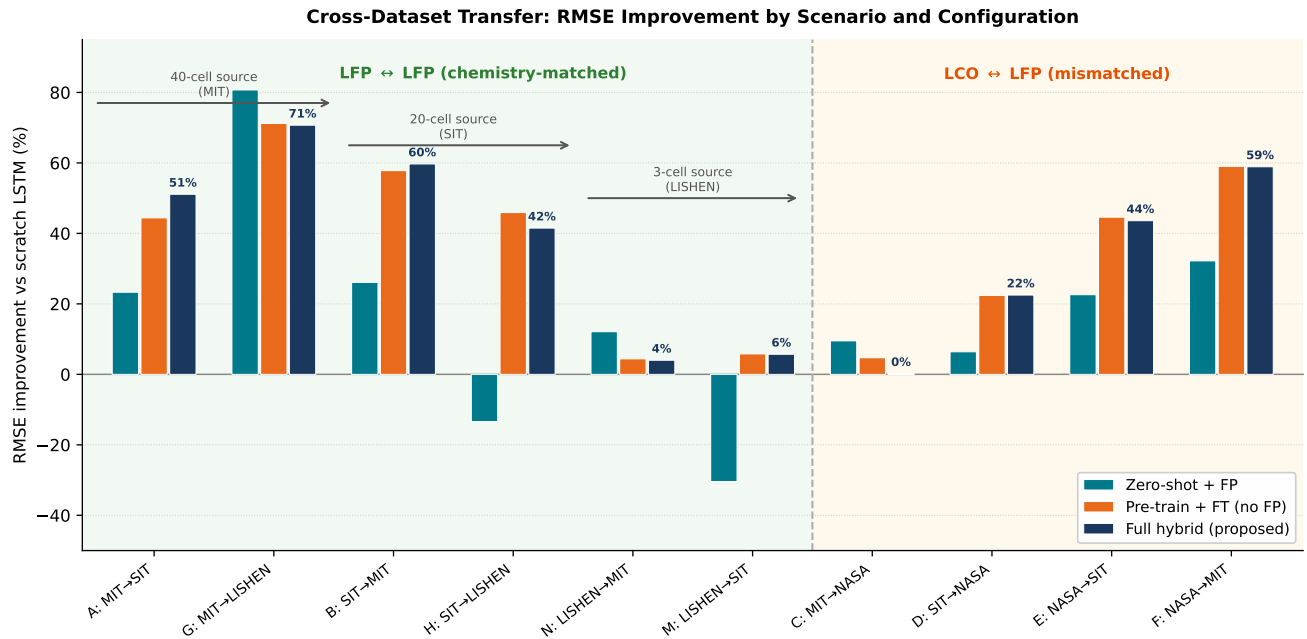
7.2.3. Chemistry mismatch reduces accuracy gains and destroys calibration

Across the four cross-chemistry scenarios (C to F), hybrid improvement ranges from 0.2% to 58.9% (mean 31.3%),

Table 13

Cross-dataset transfer matrix: RMSE improvement over scratch LSTM.

Sc.	Source	Target	Chem.	ZS+FP	PT+FT	Hybrid
<i>LFP↔LFP (chemistry matched)</i>						
A	MIT (40 cells)	SIT	LFP	+23.3%	+44.4%	+51.1%
G	MIT (40 cells)	LISHEN	LFP	+80.7%	+71.2%	+70.7%
B	SIT (20 cells)	MIT	LFP	+26.1%	+57.8%	+59.7%
H	SIT (20 cells)	LISHEN	LFP	−13.4%	+45.9%	+41.5%
N	LISHEN (3 cells)	MIT	LFP	+12.1%	+4.4%	+4.0%
M	LISHEN (3 cells)	SIT	LFP	−30.4%	+5.8%	+5.7%
<i>LCO↔LFP (chemistry-mismatched)</i>						
C	MIT (LFP)	NASA (LCO)	cross	+9.5%	+4.7%	+0.2%
D	SIT (LFP)	NASA (LCO)	cross	+6.4%	+22.4%	+22.5%
E	NASA (LCO)	SIT (LFP)	cross	+22.6%	+44.6%	+43.6%
F	NASA (LCO)	MIT (LFP)	cross	+32.2%	+59.0%	+58.9%

**Figure 13:** Cross-dataset RMSE improvement over scratch LSTM across all ten transfer scenarios.

against 41.5% to 70.7% (mean 55.7%) for the adequately resourced matched-LFP scenarios (A, B, G, H). The sharper signal is calibration: the PI coverage analysis from Section 7.1 (80.8% LFP→LFP vs 43.6% LFP→LCO) shows that the RMSE improvement metric understates the damage from chemistry mismatch. The model can remain reasonably accurate in point estimates while becoming severely overconfident in its uncertainty intervals, the failure mode most dangerous for BMS deployment, and one invisible to every uncertainty-free method in Table 5.

7.3. Deployment-Realistic Evaluation

The 30/50/70% splits of Section 4.3 are retrospective: each presupposes the cell's full lifetime. To match how the model is actually used, we re-evaluate under fixed early-life budgets, fine-tuning on only the first 30, 50, 100, or 200 cycles and predicting the remainder. Table 14 shows the hybrid improves over the equally-budgeted scratch LSTM at every budget, by 13% to 37% mean per-cell RMSE on the 17 test cells. The gain is present already at the smallest budget, a 37% reduction from only the first 30 cycles, which is the regime that matters for a newly commissioned cell with no usable history and where a retrospective split would not apply.

Table 14
Fixed early-life budget evaluation.

Budget (cycles)	Hybrid RMSE	Scratch RMSE	Improvement
30	0.031	0.045	37.1%
50	0.031	0.043	34.0%
100	0.031	0.043	13.3%
200	0.024	0.040	35.7%

7.3.1. Intermittent SOH observation

The one-step results assume the true SOH is available at every cycle. In a battery management system the discharge capacity, and hence SOH, may be estimated only intermittently. We therefore supply the observed SOH only every k cycles and let the model feed its own prediction back between observations. Mean one-step RMSE over the 17 test cells rises from 0.015 at $k = 1$ to 0.055, 0.114, and 0.202 at $k = 5, 10$, and 25. The near-persistence accuracy of the one-step benchmark therefore holds only under frequent SOH estimation; as the observation gap widens the errors compound through the self-fed window. This is a further reason the one-step metric is not the right basis for judging deployment value, and it motivates the trajectory-level remaining-useful-life forecaster of Section 9, which predicts from a single early-life anchor and does not require per-cycle SOH.

8. Early Detection of Manufacturing-Defective Cells

Section 5 established that cell-to-cell variability dominates within the G1 group, where nominally identical cells

span a thirteenfold range in final SOH. A practical question follows. Can the defective cells be identified early, before their capacity has visibly diverged? We find that they can, but only from the shape of the discharge voltage curve, not from any scalar health indicator.

8.1. Scalar indicators are blind within a group

Within G1, the early-life value and slope of the discharge capacity, the surface-temperature resistance proxy, the coulombic efficiency, and the DC internal resistance all fail to separate the fast-failing cells from the robust ones. Every within-group rank correlation with final SOH is weak and not significant, with $|\rho| < 0.32$ and $p > 0.2$. In other words, at fixed operating condition the standard health indicators carry no early warning of which cell will fail.

8.2. The voltage curve carries an early signature

The discriminative signal instead lies in how the discharge capacity voltage curve evolves. We use the Severson descriptor, the change in the curve between an early reference cycle and a later screening cycle, $\Delta Q(V) = Q_j(V) - Q_i(V)$ [18], summarised by $\log_{10} \text{Var } \Delta Q(V)$. Evaluated at a screening cycle of 100, this scalar classifies eventual early death, defined as final SOH below 0.80, at a leave-one-out AUC of 0.78, well above the 0.64 obtained from the scalar capacity value alone in Section 5. The same descriptor and cutoff reproduce on the independent MIT LFP corpus, at AUC 0.75 and with a Pearson correlation of -0.88 between $\log_{10} \text{Var } \Delta Q(V)$ and log cycle life. This gives a usable screen by cycle 100 on both a large-format and a small-format LFP dataset.

8.3. How early is the warning

The distinctive value of the voltage signature is that it appears before the capacity signal. Figure 14 plots the detection AUC against the screening cycle for a classifier built on the voltage descriptor and for one built on the observed capacity fade trajectory. At cycle 50 the capacity trajectory carries no usable prospective signal: with so few cycles the fitted fade-slope direction does not generalise across cells and its leave-one-out AUC collapses toward zero, whereas the SIT voltage descriptor already discriminates at an AUC of 0.65, the only signal above the scalar capacity ceiling at that point. By cycle 75 the capacity trajectory has caught up, and by cycle 100 it slightly exceeds the voltage descriptor. The MIT cross-validation follows the same pattern, becoming reliable only by cycle 100. We therefore make a deliberately bounded claim. The voltage descriptor buys roughly 50 cycles of lead time over the capacity trajectory on the SIT cells, identifying which cells will fail before the capacity fade reveals it; once enough cycles accumulate, the capacity trajectory itself suffices. The descriptor does not by itself sharpen the prediction of when or how far a given cell will collapse. That regression task remains bounded by the difficulty documented in Section 9.4.

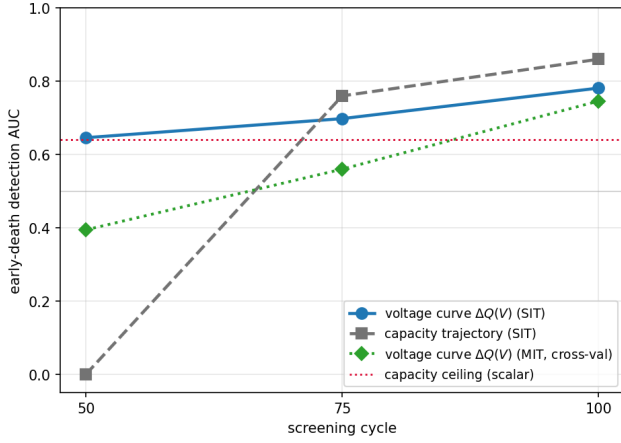


Figure 14: Early detection AUC versus screening cycle.

9. Remaining Useful Life Forecasting

The hybrid of Section 6 predicts the next-cycle SOH. For deployment the decision-relevant quantity is the remaining useful life (RUL), defined here as the number of cycles until a cell first reaches the 0.80 end-of-life (EOL) threshold. Converting a one-step model into an RUL forecaster is not trivial. The natural approach is to roll the model forward autoregressively, feeding each prediction back as the next input. We find, consistent with recent forecasting studies [33, 34], that this recursive scheme accumulates error over the horizon and, once a cell enters nonlinear accelerated ageing, tends to extrapolate the last observed near-linear slope and miss the knee. On the SIT cells this failure is severe. The recursive lineage forecasts an RUL for only 38% to 86% of the cells that reach EOL, and it abstains on the very cells that matter, the fast-failing G1 cells 001-8 and 101-3 that carry the early-life defect.

Evaluation population and censoring. Of the 20 SIT cells, 12 reach the 0.80 EOL threshold within the observation window and 8 do not. The latter eight (001-2, 001-5, 001-6, 001-7, 002-1, 002-2, 002-3, and 002-5) are right-censored: each still holds an SOH between 0.85 and 0.92 at the end of its record, so its true RUL is known only to exceed the observed span. All RUL error statistics in this section are therefore computed on the 12 cells that reach EOL, since a censored cell has no ground-truth RUL against which to score a forecast. The foundation model nonetheless returns a forecast for every censored cell, in each case placing the first 0.80 crossing 190 to 324 cycles beyond the forecasting anchor, consistent with their shallow fade, and never inside the already-observed span.

The same convention applies to every method in the comparison of Section 9.4: a censored cell carries no ground-truth RUL, so it enters no error statistic for any of them, and the coverage column of Table 15 is computed over the 12 cells that reach EOL. What differs is what each method emits. The direct decoder returns a crossing for all eight censored cells, as above. The shortfall in the recursive lineage’s coverage arises within the EOL set, on the catastrophic cells,

rather than on the censored ones. We did not separately characterise how the recursive rollouts behave beyond the end of a censored cell’s observed record, and we note that as a limitation rather than claiming a result we did not measure.

9.1. Direct trajectory decoding

We extend the hybrid with a direct multi-horizon decoder that predicts the whole remaining SOH trajectory in a single forward pass, which removes the iterative feedback and its error accumulation [33, 35]. The pretrained encoder, comprising the early-life fingerprint and the observed window, is retained unchanged. A compact decoder head outputs the predicted SOH at 14 fixed horizons, namely 10, 20, 30, 40, 50, 75, 100, 150, 200, 250, 300, 400, 500, and 600 cycles ahead of the forecasting anchor, and the RUL is read as the first horizon at which the predicted curve reaches 0.80.

Decoder specification. The head takes the concatenation of the LSTM encoding of the last $W = 20$ observed cycles and the 16-dimensional early-life fingerprint, passes it through one hidden layer of 64 ReLU units with dropout 0.2, and emits the 14 horizon values through a linear layer. It is trained with a plain mean-squared-error loss applied equally across the 14 horizons, with no horizon-dependent weighting, at an Adam learning rate of 10^{-3} . Training targets are the observed SOH at each horizon, with horizons that extend past a cell’s record padded by its final observed value. At inference the crossing cycle is obtained by linear interpolation between the two horizons that bracket 0.80; when no horizon reaches 0.80 the model reports no crossing rather than extrapolating a slope. We do not enforce monotonic degradation: each horizon is predicted independently and the curve is only clipped to the interval 0 to 1.2, so a nonmonotone prediction is admissible, though it did not affect the reported crossings. Uncertainty follows from 30 Monte Carlo Dropout passes; the point RUL is read from the median curve and the 80% interval from the 10th and 90th percentiles of the per-pass crossings, reported only when at least a third of the passes produce a crossing.

We keep the decoder deliberately simple. On the 20-cell target set, richer architectures overfit and lose coverage: an encoder-decoder LSTM with attention reaches an RUL error of 96 and 43 cycles at the 30% and 50% anchors with coverage 79%, and a transformer self-attention encoder reaches 95 and 72 cycles, whereas the compact decoder reaches 68 and 49 cycles at full coverage. This is consistent with reports that simpler models dominate when target data are scarce [36].

9.2. Foundation pretraining

Because the target set is small, transfer is decisive. We extend the single-source pretraining of the base hybrid into a multi-dataset LFP foundation corpus of 155 cells drawn from the MIT, HUST, and SNL collections, and we pretrain the trajectory decoder on this corpus before leave-one-cell-out fine-tuning on the SIT cells. This is the largest single lever we found. Table 15 reports a clear ablation ladder. Training from scratch gives 182 and 102 cycles of error at the 30% and 50% anchors, MIT-only pretraining gives 85

and 100, and the 155-cell foundation corpus gives 68 and 49. Foundation pretraining also repairs calibration. The from-scratch model is overconfident, with prediction interval coverage of 9% to 40%, whereas the foundation model attains 82% to 100%.

9.3. Probabilistic RUL

We propagate the Monte Carlo Dropout ensemble through the trajectory to obtain a predictive RUL distribution, and we report the median together with an 80% interval calibrated by leave-one-out conformal inference [37]. The calibrated intervals attain 80% empirical coverage, and at the 50% anchor the median width tightens from 285 to 175 cycles relative to the raw ensemble. At the 30% anchor the intervals remain wide, which honestly reflects the intrinsic difficulty of forecasting a thirteenfold spread in end-of-life from early data.

9.4. Forecasting Accuracy and Coverage

Table 15 and Figure 15 summarise the comparison against the recursive lineage; errors are in cycles, and a blank prediction-interval-coverage entry marks methods that produce no intervals. The foundation model forecasts every cell that reaches EOL, including the two catastrophic cells the recursive methods cannot, at a median error of 68 and 49 cycles and with calibrated intervals. The recursive DE-LSTM attains a lower error, 41 and 12 cycles, but only on the easy 38% to 86% of cells that it chooses to forecast. This is the coverage against risk trade-off familiar from selective prediction [38]. A method that withholds a forecast on the defective cells is of limited use for a manufacturing screen, where those cells are precisely the ones of interest.

Figure 16 plots the two quantities against each other so that neither can be read in isolation. Accuracy alone would rank the recursive lineage first at the 50% anchor, where DE-LSTM and the autoregressive hybrid reach 12 and 11 cycles against our 49. Coverage alone would rank them last. Read together, the lineage occupies the low-error but incomplete-coverage region, and it reaches that position by declining exactly the cells a manufacturer needs a forecast for; our model is the only one at full coverage at both anchors. We state the comparison this way rather than claiming a lower error than every baseline, because on the cells they do forecast at the 50% anchor the recursive methods are genuinely the more accurate. The direct decoder also reduces the error of the autoregressive hybrid from 98 to 68 cycles at the earliest anchor.

10. Discussion

This section interprets the results for deployment, covering manufacturing variability, the saturation of the standard one-step benchmark, the chemistry alignment effect, the complementary roles of the early-life fingerprint and fine tuning, pretraining-run variance and reproducibility, and the model's computational footprint, before setting out the study's limitations.

Table 15

RUL forecasting on the SIT cells.

Anchor	Method	Cov.	Med. err.	PI cov.
30%	Hybrid, foundation	100%	68	82%
	abl. MIT only	100%	85	
	abl. from scratch	100%	182	9 to 40%
	DE-LSTM, recursive	38%	41	
	GA-LSTM, recursive	38%	78	
50%	Hybrid, recursive	62%	98	
	Hybrid, foundation	100%	49	90%
	abl. MIT only	100%	100	
	DE-LSTM, recursive	86%	12	
	Hybrid, recursive	86%	11	

10.1. Cell-to-Cell Variability as the Limiting Factor

The central lesson of the SIT data is that prediction error lives in the cell, not the algorithm: within-cell trajectories are highly predictable, yet nominally identical cells diverge so widely (Section 5.2) that cell identity, not the choice of method, sets the error floor (Section 5.1). The field's effort, dominated by architectural innovation, is thus allocated against where the variance actually lies. For a BMS engineer the practical consequence is an asymmetric risk: same-cell models underestimate catastrophic degraders, while naive cross-cell pooling degrades ordinary robust cells; the fingerprint encoder is our response, conditioning each prediction on the cell's own early-life behaviour before any fine tuning begins. The same variability also explains why benchmarks built on hand-picked, smoothly degrading cells are systematically optimistic, and why a dataset that spans catastrophic to exceptionally robust cells is a more honest proving ground.

10.2. Benchmark Saturation: A Floor Check, Not a Leaderboard

The most consequential methodological finding is that rolling one-step-ahead prediction is saturated on these smooth trajectories: a trivial 20-cycle linear autoregressor matches or beats every deep model, and inter-method gaps fall within the run-to-run variance of training itself (Section 5.1). The obvious harder alternative, recursive multi-step rollout, discriminates no better here, since feeding predictions back accumulates error until every model is unusable. One-step RMSE is therefore a floor check, not a leaderboard. We accordingly recommend that studies on smooth-degradation data (i) always include a trivial persistence or linear baseline, (ii) report multi-run variance under deterministic seeding, and (iii) rank methods on the capabilities the application needs, early-life prediction, calibrated uncertainty, and cross-chemistry robustness, rather than on fourth-decimal RMSE. The same discipline applies across operating conditions: apparent temperature-transfer effects here reflect test-set difficulty rather than easy transfer, and must be controlled for (Section 5.3).

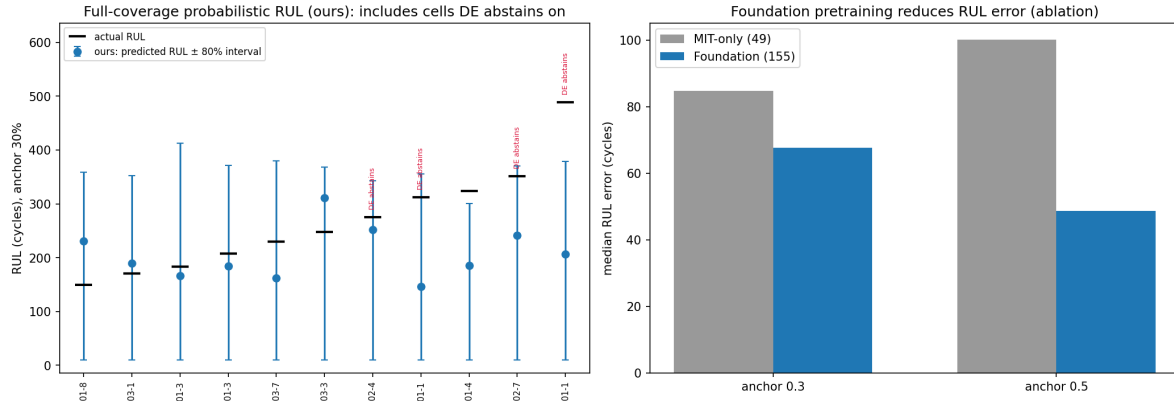


Figure 15: Full-coverage probabilistic RUL and the pretraining ablation.

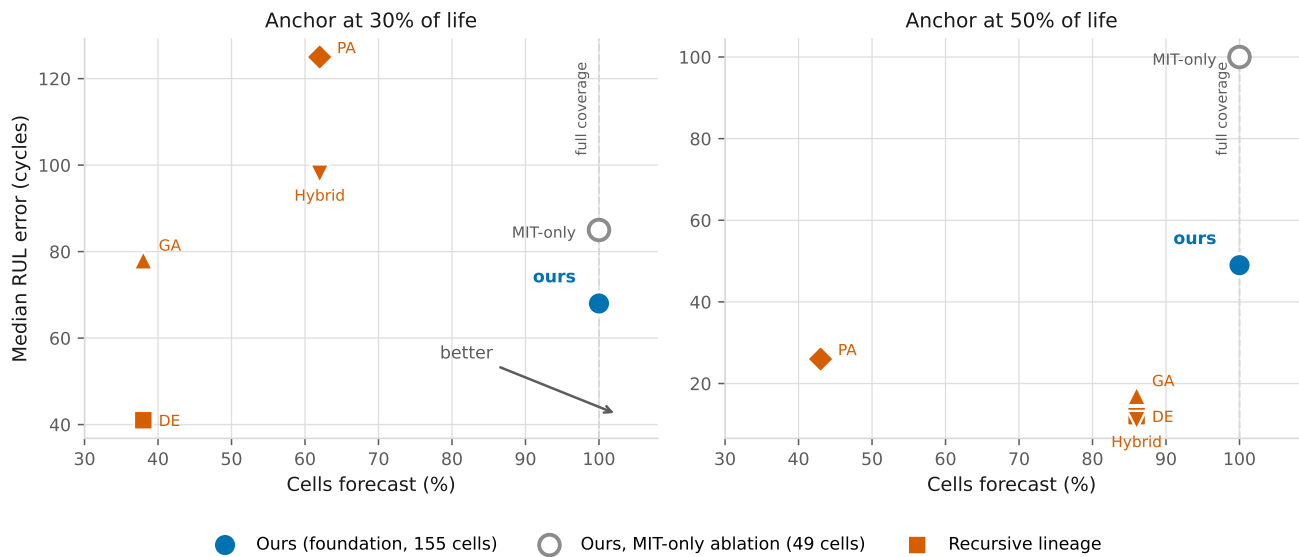


Figure 16: Risk versus coverage for RUL forecasting.

10.3. The Chemistry and Domain Alignment Effect

Uncertainty calibration transferred only when the pre-training and deployment chemistries matched (Section 7.2). The mechanism is simple: the model's learned sense of how uncertain to be is tuned to LFP degradation dynamics, so its intervals are appropriately scaled on LFP cells but overconfident on the faster-fading LCO cells. The practical rule is to pretrain on chemistry-matched data. Because even matched intervals under-cover, we regard split-conformal correction (or temperature scaling [39]) as mandatory before prediction intervals are trusted in a BMS, whether or not the chemistry is matched.

10.4. Fingerprint and Fine Tuning: Distinct Roles, Not Rivals

The ablation and its shuffled-fingerprint control together show that the fingerprint's role is to anchor the LSTM

in the pretrained population's regime rather than to identify the individual cell, a weaker mechanism than we first hypothesised but a more robust one, since it needs only a representative pretraining population, not 30 cycles rich enough to tell cells apart. The cross-dataset experiment then separates the two levers by role: fine tuning is the *dependable* mechanism, positive in every transfer scenario including each cross-chemistry pair, whereas fingerprint conditioning without fine-tuning is the *early-life* mechanism, the only option before any history exists and most reliable exactly when chemistry and corpus breadth match, the setting a manufacturer qualifying a product line can guarantee. For a BMS this means a chemistry-matched model is useful from the first cycles at no adaptation cost, with fine tuning taking over as history accumulates and becoming indispensable once chemistries differ.

10.5. Pretraining-Run Variance and Reproducibility

A surprising hazard is how strongly transfer quality depends on the pretraining draw, and that no source-domain metric we tested identified the poorly transferring runs (Section 6.3); one-step holdout RMSE was in fact inverted, rewarding collapsed persistence-like solutions. Any transfer pipeline whose pretraining objective admits a trivial shortcut may share this silent failure mode, which single-run studies cannot detect; our calibration-cell protocol is a deployable safeguard, and we recommend that battery-prognostics transfer studies report multi-run variance. The companion discipline is determinism: every number in this paper reproduces byte-for-byte from a fresh clone (Section 6.6), a practice we adopted after finding that library-default seeding alone could alter experimental conclusions.

10.6. Computational Footprint

A recurring justification for this work is BMS deployment, which constrains model size and inference cost, so we report both. The hybrid is lightweight: 21 089 parameters, approximately 82 KB in float32 and 41 KB in float16, small enough to reside in the memory of an embedded BMS microcontroller. A point SOH estimate is a single forward pass; the 50-pass MC-Dropout uncertainty estimate, when the passes are evaluated as one batched forward (replicating the input across the batch dimension), completes in about 4 ms on both a desktop CPU and the RTX 2080 Ti. Because SOH is updated once per charge-discharge cycle (tens of minutes to hours), this leaves a latency margin of several orders of magnitude; even an unoptimised implementation that runs the 50 passes sequentially (≈ 1 s) would be comfortably within budget. These figures were measured on a desktop CPU rather than production embedded silicon, so absolute timings would differ on a microcontroller; nonetheless the 41 KB footprint and few-millisecond batched inference indicate the model fits well within the resource envelope of contemporary BMS hardware.

10.7. Limitations

The findings reported above should be read alongside the boundaries of the present study. The limitations listed below stem mainly from dataset scale and the scope of the experimental design rather than from the hybrid framework itself, and each points to a concrete opportunity for future work: larger cell cohorts would sharpen the statistical claims, shorter fingerprint windows and format-matched pretraining data would broaden applicability, and post hoc calibration would close the residual coverage gap. We state them explicitly so that practitioners can judge how far the present conclusions transfer to their own deployment setting.

1. *G1 sample size*: With ten G1 cells, LOOCV results have limited statistical power. The observed two-sided pattern (pooling rescues catastrophic degraders and harms robust cells) is directionally consistent, but quantitative conclusions should be interpreted with this sample size in mind.

2. *Fingerprint window*: The 30-cycle fingerprint requires at least 30 discharge cycles before cell-specific predictions are available. For applications requiring earlier prediction (e.g., first 10 cycles), shorter window studies would be needed.
3. *Residual calibration gap*: grouped conformal recalibration restores nominal coverage but only conservatively, since just sixteen calibration cells make the finite-sample correction wide, and two extreme cells (001-8, 003-5) remain under-covered because they violate the exchangeability that conformal prediction assumes. Conformal guarantees assume exchangeability between calibration and test cells, which manufacturing outliers violate by definition; detecting such cells online (e.g. from interval-width anomalies) remains open.
4. *Format mismatch*: The MIT pretraining cells are 18650 cylindrical (1.1 Ah); the SIT cells are 50 Ah prismatic. Despite this format difference, the LFP chemistry match appears sufficient for meaningful transfer. Format-matched pretraining data would likely reduce the remaining gap further.
5. *LISHEN source corpus size*: The LISHEN dataset contributes only three cells to the cross-dataset evaluation (scenarios M and N). The consistently weak gains in these scenarios (-24.6% to $+14.6\%$ for ZS+FP) are attributed to insufficient source diversity rather than a fundamental limitation of the fingerprint mechanism. Future work should re-evaluate with a larger LISHEN corpus.
6. *Calibration-cell requirement*: The checkpoint-selection protocol requires three target-domain cells cycled to (or near) end of life. This is realistic for a manufacturer qualifying a product, but not for retrofitting arbitrary third-party packs; in that setting a chemistry-matched public dataset could serve as a weaker substitute, which we have not evaluated.
7. *Comparison scope*: Beyond the four LSTM-family predictors (Simple LSTM, PA-LSTM, GA-LSTM, DE-LSTM), we benchmarked two modern non-recurrent architectures (TCN, Transformer) and two trivial baselines under the identical protocol (Table 5). The modern baselines were run under fixed configurations rather than the per-cell hyperparameter search afforded to GA/PA/DE-LSTM; given the saturation of the one-step metric we do not expect tuning to change the conclusions, but other model classes (e.g. Gaussian-process regression) remain for future work.
8. *Evaluation protocol*: The saturation finding implies that one-step RMSE, including our own accuracy numbers, has limited power to rank methods on these data. Harder protocols (fixed early-life budgets, multi-step horizons with scheduled sampling, knee-onset prediction) are the right arena for future accuracy claims; we did not re-benchmark all methods under such protocols here.

11. Conclusion

We investigated the deployment challenges for battery SOH prediction that are absent from small-cell laboratory benchmarks, using a new 18-month dataset of 20 large-format LiFePO₄ prismatic cells (50 Ah) cycled under three operating conditions.

Its central message is that on large-format cells the cell identity, not algorithm choice, explains most of the prediction-error variation, and that because one-step accuracy on their smooth trajectories saturates, models should be judged by the deployment capabilities they add rather than by one-step rank. We therefore report trivial baselines and multi-run variance as standard practice.

Guided by this, our hybrid model (MIT LFP pretraining, a convolutional fingerprint encoder, and MC-Dropout uncertainty) is positioned not as the most accurate one-step predictor but as the one that supplies what deployment needs: prediction of a new cell from only its first 30 cycles with no target-cell training, a substantial accuracy gain once fine tuning begins, and calibrated uncertainty after a lightweight conformal correction. A ten-scenario cross-dataset validation shows that fine tuning is the dependable adaptation mechanism, that fingerprint conditioning without target-cell fine-tuning adds distinctive value when chemistry and corpus match, and that uncertainty stays calibrated only when pretraining and deployment chemistries match, a chemistry-alignment effect that offers a practical guideline for BMS transfer learning. We also identify pretraining-run variance as a silent reproducibility hazard and neutralise it with a calibration-cell selection protocol, in a fully deterministic, byte-for-byte reproducible pipeline.

Building on the same model, we demonstrated two further capabilities aimed directly at cell-to-cell variability: an early screen that flags defective cells from the discharge voltage curve well before the capacity trajectory reveals them, and a full-coverage probabilistic RUL forecaster that, through a direct multi-horizon decoder and a 155-cell LFP foundation corpus, forecasts every cell reaching end of life, including the catastrophic cells recursive baselines abstain on.

Taken together, this work moves SOH and RUL prediction beyond the small-cell laboratory setting that has dominated the field, contributing a large-format LFP benchmark with trivial-baseline controls, evidence that the field's default evaluation protocol needs rethinking, an uncertainty-calibrated and deployment-oriented model validated across manufacturers, chemistries, and operating conditions, and a fully deterministic pipeline with a released checkpoint. Future work should pursue harder evaluation protocols on which deep models may separate from trivial baselines, shorter fingerprint windows and format-matched pretraining, online detection of cells that violate conformal exchangeability, and pretraining objectives that exclude trivial shortcut solutions.

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Data and Code Availability Statement

The SIT large-format LiFePO₄ dataset analysed in this study is openly available at <https://doi.org/10.25447/sit.32101523> [10]. The MIT/Severson fast-charging dataset is openly available from Severson et al. [18], and the NASA dataset from the NASA Prognostics Center of Excellence [6]. This study is designed to be fully reproducible. All code, the complete experimental pipeline, and the selected pre-trained checkpoint that generate every table and figure in this paper are available at <https://github.com/karthicgrepo/battery-soh-prediction>. Every experiment runs deterministically, using TensorFlow op-determinism with full float32 precision and complete seeding of the Python, NumPy, and TensorFlow random number generators, so each reported value reproduces bit-for-bit on the same hardware and software stack.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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CRedit authorship contribution statement

Karthickumar Ponnambalam: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing - original draft. **Sivaneasan Balakrishnan:** Resources, Data curation, Supervision, Writing - review and editing. **Anurag Sharma:** Supervision, Validation, Writing - review and editing. **Sze Sing Lee:** Supervision, Writing - review and editing.

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